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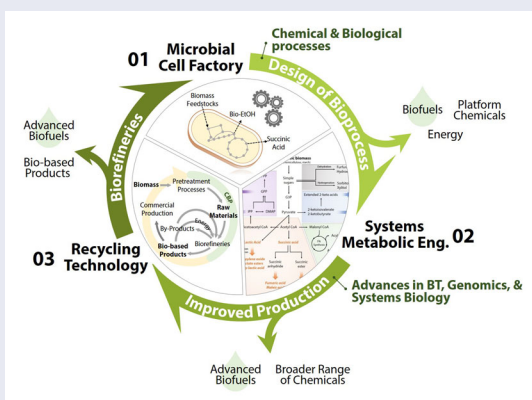
Current status and future prospects of biological routes to bio-based products using raw materials, wastes, and residues as renewable resources

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ABSTRACT

Current fossil-based commercial products pose a serious threat to global reserves of natural resources and the preservation of the natural environment. During recent decades, great efforts have been made to increase the availability of non-utilizable biomass as alternative feedstocks and reduce environmental pollution to achieve a more sustainable bioeconomy. Several bio-sectors have emerged for the production of bio-based products to replace fossil-based equivalents through bioprocesses using biomass feedstocks. Indeed, advanced microbial fermentation technologies encompassing metabolic engineering and genome-based systems biology approaches have enabled the design and development of new bio-based refineries using engineered platform cells. Herein, we focus on recent progress in the area of microbial fermentation-aided bioprocesses for the production of bio-based products derived from naturally occurring biomasses as feedstocks. Furthermore, we discuss the application of bio-based products and remaining technical barriers and assess possible biorefineries.



Abbreviations: 1G: First-generation; 2G: Second-generation; 3G: Third-generation; AD: Anaerobic digestion; BDO: Butanediol; BioDME: Bio-dimethylether; BioSNG: Bio-synthetic natural gas; BtL: Biomass to liquids; CBP: Consolidated bioprocessing; CHP: Combined heat and power; DME: Dimethylether; EU-28: the 28 countries of the EU; FAEE: Fatty acid ethyl esters; FAME: Fatty acid methyl esters; FOS: Fructo-oligosaccharides; GHG: Greenhouse gas; HEFA: Hydroprocessed esters and fatty acids; HHV: High heating value; HMF: 5-Hydroxymethylfurfural; HTC: Hydrothermal carbonization; LA: Lactic acid; LCB: Lignocellulosic biomass; LTP: Low-temperature pyrolysis; MEG: Monoethylene glycol; MSW: Municipal solid waste; MTBE: Methyl tert-butyl ether; PBS: Polybutylene succinate; PBT: Polybutylene terephthalate; PDO: Propanediol; PE: Polyethylene; PET: Polyethylene terephthalate; PG: Propylene glycol; PHA: Polyhydroxyalkanoate; PLA: Polylactic acid; PP: Polypropylene; PTT: Polytrimethylene terephthalate; PVC: Polyvinyl chloride; SA:

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Succinic acid; TAME: Tert-amyl methyl ether; THF: Tetrahydrofuran; TOE: Tons of oil equivalent; WP: Wood pellet

KEYWORDS Bio-based products; biomass; biorefinery

1. Introduction

The global human population is expected to rise to 9.15 billion by 2050 (Alexandratos & Bruinsma, 2012). Accordingly, increased global demand for natural resources will far exceed their ability to support industry, agriculture, and fisheries, especially food, clean water, energy, animal feed, and materials for clothing and housing, as well as other uses (FAO, 2011; Shafiee & Topal, 2009). Due to uncertainty over the availability of natural resources, the production, use, and recycling of biomass using sustainable and renewable systems is also required (Ragauskas et al., 2006; Speirs et al., 2015). Moreover, current fossil fuel-based consumption and production patterns pose a serious and irreversible threat to the environment (Bengtsson et al., 2018; Pacala & Socolow, 2004). Currently, we are facing new challenges to reduce greenhouse gas (GHG) emission, provide clean energy, and secure public health (Owusu & Asumadu-Sarkodie, 2016). Urgent circumstances and plausible scenarios compel us to design and develop bio-based processes and products to meet higher environmental and industrial standards. Therefore, we must find sustainable alternative resources and develop practically feasible ways for their usability to balance biomass supply and demand, and thereby build a successful bioeconomy (Goldemberg, 2007; Piotrowski et al., 2016).

Regarding the future of sustainable bio-based industries, including supply and demand for biomass, there remain technical barriers to overcome, and clear objectives should focus on producing cost-effective bio-based products that perform better than conventional, fossil-based products. In this regard, we must utilize renewable resources as sustainable biomass feedstocks that can be valorized for bio-based production, reduce petroleum dependency, diversify feedstocks, and develop novel recyclable and compostable materials (Andler & Goddard, 2018; Brehmer, 2008; Clark et al., 2015; Malico et al., 2019). Moreover, the production of energy and high-value biorefinery products is paramount, as is the preservation of biodiversity, clean water, fresh air, soil fertility, and other environmental factors. The development of advanced technologies for replacing depleted nonrenewable resources (e.g., oils, coal, minerals, and natural gas) is also important. When these prerequisites are satisfied, the development of bio-based products, including food, platform chemicals, and sustainable energy from renewable sources, will

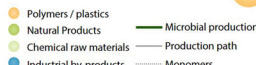
enable us to shift away from our current heavy use of fossil fuel resources. In this review, we mainly address the current status of biological routes to the production of bio-based products from biomass feedstocks and discuss their future contributions toward a sustainable bioeconomy and minimization of environmental deterioration while optimizing the use of raw materials, by-products, and waste. We also discuss recent advances in bio-products using engineered microorganisms for bioprospecting.

2. Biomass feedstocks and bio-based production processes

2.1. Feedstocks

Recently, a variety of organic residues including biological wastes have been implemented to serve as feedstocks in biorefinery processes for the production of low carbon products, functional compounds, and biofuels (Schieber et al., 2001; Tripathi et al., 2019). Potential biomass feedstocks originate from agricultural, forestry, and industrial activities; some are from dedicated crops specifically produced as exploitable biomass, and industrial organic wastes and by-products (Salassi et al., 2017; Sands et al., 2017). Cellulose, starch, sugars, hemicellulose, lignin, oils, proteins, ashes, and various chemical elements (e.g., CO₂, H₂, S, and N) are the basic components of biomass feedstocks (Figure 1). In general, biomass can be classified into four groups; lignocellulosic biomass (LCB), forestry biomass, agro-based biomass, and waste biomass.

LCB mainly originates from forest residues, and is an abundant potential feedstock for many bio-based products and biofuels (Bhutto et al., 2016; Hood, 2016). These fibrous, non-edible plant materials can be converted to various chemicals, advanced biofuel compounds, and other functional molecules (Brethauer & Studer, 2014; Saini et al., 2015; Soccol et al., 2019) (Figure 1). Thus, the future of dedicated lignocellulosic cropping is promising because LCB does not compete with human food or animal feed, or dedicated energy crops (Passoth & Sandgren, 2019). Despite the attractiveness of LCB as an alternative to fossil-based resources, there are several practical drawbacks, such as inhibition of enzymatic hydrolysis, poor cellulose digestibility, the release of toxic chemicals into the environment, and high energy demand (Capolupo & Faraco, 2016). Furthermore, the amount of lignin, the chain length, and the degree of polymerization can vary depending on plant species, resulting in heterogeneity of many lignocellulosic wastes and residues, accompanied by challenges to logistics, handling, and market development (Welker et al., 2015). To overcome these technical and economic barriers, many research groups have attempted to maximize the potential of LCB for the production of various bio-based products using a consolidated bioprocessing (CBP) strategy and integrated



ished products (adapted from <https://www.kumu.io/alcn/c-mfa-o#bio-based>). 1,2-PDO, 1,2-Propanediol; 1,3-PDO, 1,3-Propanediol; 1,4-BDO, 1,4-butanediol; 2,3-BDO, 2,3-Butanediol; 3-HPA, 3-Hydroxypropionic acid; P(3-HPA), Poly (3-hydroxypropionic acid); DFF, 2,5-Diformylfuran; DMDA, 1,10-decamethylene; DMF, Dimethylformamide; DMI, dimethylisorbitol; FAME, fatty acid methyl ester; FDCA, 2,5-Furandicarboxylic acid; 5-HMF, hydroxymethylfurfural; LDI, Lysine diisocyanate; MEA, ethanolamine; MEG, Monoethylene glycol; PA, Polyamide; PAA, Polyacrylic acid; PBF, Polyethylene 2,5-furandicarboxylate; PBS, Polybutylensuccinate; PBSA, polybutylensuccinatadipate; PBT, Polybutylene terephthalate; PE, Polyethylene; PEF, Polyethylene furanoate; PET, Polyethylene terephthalate; PHA, Polyhydroxyalkonate; PHB, Polyhydroxybutyrate; PLA, Polylactic acid; PMMA, Polymethyl methacrylate; PP, Polypropylene; PS, Polystyrene; PTT, Polymethylene terephthalate; PUR; Polyurethane; PVA, Polyvinylacetate; PVC, Polyvinylchloride; SBR, Styrene-butadiene rubber; TA, Terephthalic acid; TPS, Thermoplastic strength; THF, Tetrahydrofuran.

biorefineries with engineered microbial platforms (see Sections 2.2 and 2.3) (den Haan et al., 2015; Gowen & Fong, 2011; Xu et al., 2009). Nevertheless, recent advances in thermochemical technologies will likely increase the market demand for dedicated lignocellulosic crops in future years to support advanced biofuels production (Table 1).

Forestry biomass, mainly lignin, is currently exploited to produce bio-based products and bioenergy (Dessbesell et al., 2017; Karatzos et al., 2017). In the case of advanced biofuels or intermediates, bio-synthetic natural gas (BioSNG), biocrude, biomass to liquid (BtL), methanol, and bio-dimethylether (BioDME) can be produced from forestry residues and waste wood at an industrial scale (Figure 2). Recent research has focused on the diversification of biorefinery portfolios, including new value-added

Table 1. Classification of processes for the production of bio-based products.

Type	Technology	Uses	References
Mechanical/ physical processes	- Pressing - Milling - Separation - Fiber separation	Pretreatment (size reduction or separation)	(Cherubini, et al. 2009; Pap, et al. 2004)
Chemical processes	- Acid/alkaline - Wet oxidation - Green solvents	Pretreatment	(Brodeur, et al. 2011)
Biochemical processes	- Anaerobic digestion (AD) - Aerobic/anaerobic fermentation - Enzymatic conversion - Transesterification	Wet processing technique (biomass with high moisture content)	(Bacenetti, et al. 2017; Cordero-Ravelo and Schallenberg-Rodriguez 2018; Cuéllar and Webber 2008; Dimitriadis, et al. 2018; Grimberg, et al. 2015; Liao, et al. 2016; Mata-Alvarez, et al. 2014; Pereira, et al. 2015; Romero-Güiza, et al. 2016; Sikarwar, et al. 2016; Xu, et al. 2018)
Thermochemical processes	- Combustion - Gasification - Pyrolysis - Hydrothermal upgrading - Torrefaction - Liquefaction - Hydrogenation	Dry woody and herbaceous biomass	(Dessbesell, et al. 2016; Gardezi, et al. 2013; Gollakota, et al. 2018; Jahirul, et al. 2012; Karatzos, et al. 2017; Maurya, et al. 2015; Melero, et al. 2012; Sikarwar, et al. 2017; Tanger, et al. 2013; Thomas, et al. 2017; Tumuluru, et al. 2011)

chemicals from forest residues (Dessbesell et al., 2017), but biological processes have not yet been developed.

Agro-based biomass (i.e., crop stalks, and farm and crop residues) is commonly used for value-added compost (Tripathi et al., 2019). However, dedicated sugar and starch crops (e.g., wheat, barley, and maize) (Sosulski et al., 1997), including farm and crop residues, now serve as the main agro-based feedstocks for the production of bioethanol and biodiesel as first-generation (1 G) biofuels (Table 2) (Devarapalli & Atiyeh, 2015; Mohr & Raman, 2013). Wheat and corn are mainly processed to bioethanol through enzymatic hydrolysis using cellulase and hemicellulase, followed by yeast fermentation (Koti et al., 2016; Talebnia et al., 2010). Other common starch crops such as barley and rye are also employed, while maize residues are used for biogas production through anaerobic fermentation (Mazurkiewicz et al., 2019). Meanwhile, biodiesel and other bio-based platform chemicals can be produced from oil crops including rapeseed and

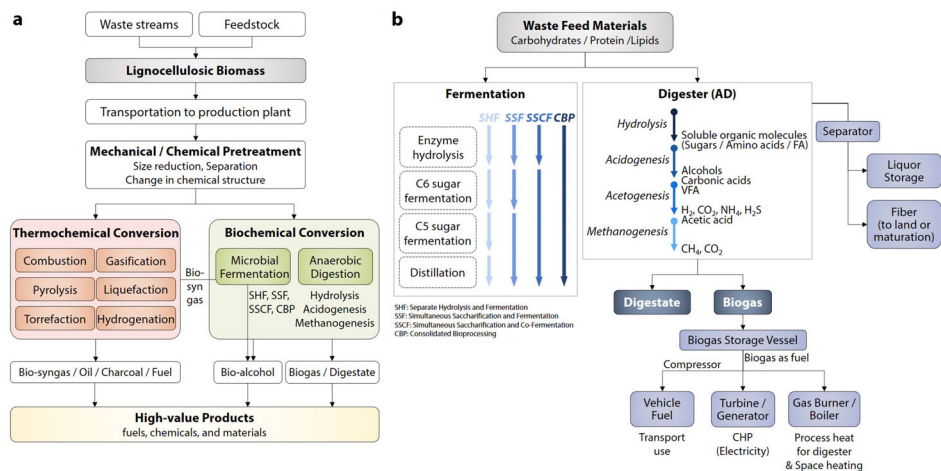


Figure 2. Conventional production of bio-based products from biomass. (a) The four major processes (mechanical, chemical, thermochemical, and biochemical). (b) Anaerobic digestion. SHF, Separate hydrolysis and fermentation; SSF, Simultaneous saccharification and fermentation; SSCF, Simultaneous saccharification and co-fermentation; CBP, Consolidated bioprocessing; FA, Fatty acids; VFA, Volatile fatty acids; CHP, Combined heat and power.

palm, coconut, soybean, and castor (Bacenetti et al., 2017). Microalgae have also been exploited for biodiesel production as a source of triacylglycerides (Delrue et al., 2012), while manure from livestock and residues from agricultural activities are mostly converted into biogas to meet local energy demands (Esteves et al., 2019; Insam et al., 2015).

There is a growing body of research focusing on exploitation of waste biomass as a potential feedstock for bio-based products using microbial fermentation. According to the Food and Agricultural Organization of the United Nations, approximately 1.3 billion tons of food waste, representing the largest proportion of municipal solid waste (MSW), is discarded globally every year (Gustavsson et al., 2011). Zero-value food waste includes rotten fruits and vegetables, offal of fish and poultry (organs, intestine, shells and bones), food leftovers, pulp and tea leaves, and dairy products. Although waste biomass does not compete with food availability or land use, conventional thermochemical decomposition of food waste is not a sustainable or eco-friendly process because it generates unwanted by-products and greenhouse gas emissions. To valorize food waste as biomass feedstock, many bioconversion processes have been developed using biocatalysts (Karmee, 2016). For example, glycerol waste, used cooking oils, and recycled vegetable oils can be converted into aviation biofuels (e.g., biokerosene and alkanes or terpenes) using native or engineered microorganisms (Table 2) (Almeida et al., 2012; Cordero-Ravelo & Schallenberg-Rodriguez, 2018). Fatty acid or alcohol biosynthetic pathways of *Escherichia coli* and *Saccharomyces cerevisiae* can be exploited for the production of

Table 2. Microbial production of the most common bio-based products and biofuels from biomass and wastes.

Product	Substrate	Microorganism ^a	Production process ^b	Titer (g/L) ^c	Yield (g/g) ^c	Prod. (g/L/h) ^c	References
Bio-based chemicals and building blocks							
Lactic Acid	Sugarcane bagasse hydrolysate	<i>Bacillus</i> sp. P38	Fermentation	185	0.99	1.93	(Peng et al., 2014)
	Corn stover hydrolysate	<i>B. coagulans</i>	CF	90	0.91	13.8	(Ma et al., 2016)
	Non-detoxified wood hydrolysate	<i>L. paracasei</i> *	FBF	99	0.96	3.23	(Kuo et al., 2015)
	Corn stover	<i>L. pentosus</i>	Fed-batch SSF	92.3	0.66	1.92	(Hu, Lin, et al., 2016)
	Softwood hydrolysate/paper mill sludge	<i>L. rhamnosus</i>	SSF	60	NR ^d	NR	(Shi et al., 2015)
Succinic Acid	Glycerol	<i>E. faecalis</i>	FBF	55.3	NR	NR	(Murakami et al., 2016)
	Cassava root	<i>A. succinogenes</i>	FBF	151.44	1.51	3.22	(Thuy et al., 2017)
	<i>Saccharina latissima</i> (microalga)		BF	36.8	0.92	3.9	(Marinho et al., 2016)
	Corn stover hydrolysate		BF	42.8	0.74	1.27	(Salvachúa, Mohagheghi, et al., 2016)
	Pretreated cane molasses		FBF	83.67	NR	1.74	(Wang et al., 2018)
	<i>Landoltia punctata</i> (duckweed)		Semi SSF	75.46	0.83	1.35	(Shen et al., 2016)
	Corn stover hydrolysate	<i>B. succiniciproducens</i>	BF	30	0.69	0.43	(Salvachúa, Smith, et al., 2016)
	<i>Arundo donax</i>		BF	17	0.75 mol/mol	0.35	(Cimini et al., 2016)
	Cassava starch	<i>E. coli</i> *	AnF	127.1	0.86	3.23	(Chen et al., 2014)
	Cassava pulp		Fed-batch SSF	98.63	0.72	1.03	(Sawisit et al., 2015)
Sorbitol Xylitol	Red algae hydrolysate		DPF	22.4	1.13 mol/mol	0.31	(Olajuyin et al., 2016)
	Sugar cane molasses	<i>Z. mobilis</i>	BF	13.87	0.046	0.38	(Cazetta et al., 2005)
	Horticultural waste	<i>C. athensensis</i>	BF	100.1	0.81	0.98	(Zhang et al., 2012)
	Corn cob	<i>C. tropicalis</i>	FBF	96.5	0.83	1.01	(Li et al., 2012)
	Hemicellulose hydrolysate	<i>E. coli</i> *	BF	150	NR	1.40	(Su et al., 2015)
	Rice straw hydrolysate	<i>S. cerevisiae</i> *	Fermentation	37.9	0.63	NR	(Guirmand et al., 2016)
		<i>C. tropicalis</i> & <i>E. coli</i> (co-production of EtOH)	CF	31	0.78	NR	(Zahed et al., 2016)
	Sugarcane bagasse	<i>C. tropicalis</i> & <i>E. coli</i> (co-production of EtOH)	FBF	24.0	0.50	NR	(Unrean & Ketsub, 2018)
	Waste xylose mother liquor	<i>C. tropicalis</i> & <i>B. subtilis</i>	SF	95	NR	0.75	(Wang et al., 2016)
Biofuels							
Biogas	Corn straw	<i>S. cerevisiae</i>	AnF	0.6 L/kg	NR	NR	(Kotarska et al., 2019)

(continued)

Table 2. Continued.

Product	Substrate	Microorganism ^a	Production process ^b	Titer (g/L) ^c	Yield (g/g) ^c	Prod. (g/L/h) ^c	References
BioEtOH	Raw glycerol	<i>C. freundii</i>	SF	16.98	NR	0.35	(Metsoviti et al., 2012)
	Microalgae	<i>Z. mobilis</i>	BF	14.5	0.45	0.7	(Ho et al., 2017)
	<i>Chlorella</i> sp. TIB-A01	<i>S. cerevisiae</i>	BF	49.1	NR	NR	(Zhou et al., 2011)
Isobutanol	Cellulose	<i>C. cellulolyticum</i> *	Fermentation	11.3	0.47	NR	(Higashide et al., 2011)
	Carbon dioxide	<i>S. elongates</i> *	SF	0.66	NR	NR	(Atsumi et al., 2009)
1,3-PDO	Raw glycerol	<i>C. freundii</i>	FBF	68.1	0.40	0.79	(Metsoviti et al., 2013)
		<i>K. pneumoniae</i>	FBF	80.2	2.29	1.16	(Jun et al., 2010)
		<i>C. butyricum</i>	FBF	61.5	NR	2.1	(Wilkins et al., 2012)
2,3-BDO		<i>E. aerogenes</i>	SF	22	NR	NR	(Metsoviti et al., 2012)
	Cassava powder	<i>E. cloacae</i>	Fed-batch SSF	78.3	0.42	3.3	(Wang et al., 2012)
FAEE	Raw glycerol	<i>S. cerevisiae</i> *	AnF	0.52	NR	0.007	(Yu et al., 2012)
Bioplastics							
PHB	Whey	<i>Caulobacter segnis</i>	FBF	9.3	NR	0.19	(Bustamante et al., 2019)
	Rice paddy straw	<i>R. eutropha</i>	Fermentation	11.42	NR	0.238	(Saratale & Oh, 2015)
	Water hyacinth hydrolysate		Fermentation	7	0.2	0.097	(Radhika & Murugesan, 2012)
mcl-PHA	Wheat straw hydrolysate	<i>B. sacchari</i>	FBF	105	0.22	1.6	(Cesário et al., 2014)
P(3HB-co-3HV)	Biodiesel-derived glycerol	<i>P. putida</i>	FBF	6.25	0.16	0.13	(Kenny et al., 2012)
Cadaverine	Miscanthus biomass	<i>R. eutropha</i>	Fermentation	2	0.14	NR	(Bhatia et al., 2019)
	Soluble starch	<i>C. glutamicum</i> *	Fermentation	23 mM	NR	NR	(Tateno et al., 2009)
Bio-based food and feed ingredients							
L-Glutamate	Rice straw hydrolysate	<i>C. glutamicum</i> *	SF	37 mM	NR	0.113	(Meiswinkel et al., 2013)
	Rice straw and wheat bran hydrolysate		SF	93 mM	NR	NR	(Gopinath et al., 2011)
L-Lysine	Rice straw and wheat bran hydrolysate		SF	42 mM	NR	0.085	
	Grass silage juice	<i>C. glutamicum</i>	BF	1.7	0.24	0.061	(Neuner et al., 2013)
	Corn silage juice		BF	1.37	0.23	0.105	
MMP-1-suppressive peptide	Poultry feathers	<i>F. islandicum</i> AW-1	AD	4.4	0.55	NR	(Jin et al., 2019)
Xanthan gum	Sugar beet molasses & chicken feather peptone	<i>X. campestris</i>	SF	20.5	0.46	0.34	(Ozdal & Başaran Kurbanoglu, 2019)
Dextran	Sugar beet molasses	<i>L. mesenteroides</i>	SF	54-55	NR	NR	(Vedyashkina et al., 2005)

bioaviation fuels from feedstock comprising lipids and carbohydrates (Wang, Dewil, et al., 2019). Likewise, animal fat waste can serve as a source of animal fatty acid methyl esters (FAMEs) for the production of biodiesel in multifeedstock production facilities (Thomas et al., 2017). In Europe, potato peel waste has great potential for the production of bioethanol through yeast fermentation (Arapoglou et al., 2010). However, the high moisture content and heterogeneous chemical composition of food waste constitutes a great challenge for their use as a biomass feedstock in energy generation. Other MSW such as commercial and residential garbage including paper, plastics, rubber, leather, and textiles can be converted into bio-fuels or biochemicals (Kayhanian, 1995; Lin et al., 2013) (Table 2). Plant oils such as tall oil and by-products from the pulp industry, palm oil refineries and the fish oil industry can also be used for biofuel production (Zahan & Kano, 2018). Recently, several pretreatment methodologies have been attempted to facilitate biorefinery approaches that maximize the conversion of organic wastes into methane (Figure 2) (Paritosh et al., 2018).

2.2. Processes

There are four main processes for the production of bio-based products derived from biomass: mechanical (or physical), chemical, biochemical, and thermochemical processes (Table 1 and Figure 2a). Mechanical or physical processes are commonly used pretreatments for biomass materials and their intermediates to reduce the size of various components and/or separate them without changing the chemical structure, whereas chemical processes may cause chemical changes in the substrate of interest (Brodeur et al., 2011; Cherubini et al., 2009). For example, physicochemical processes such as steam explosion, liquid hot water, and ammonia fiber expansion are generally used to increase the conversion yield of LCB to fermentable sugars and/or BioSNG that require cleaning and conditioning before synthesis (Maurya et al., 2015).

Thermochemical processes involving high temperature and/or pressure are widely used to convert lignocellulosic feedstock into transportation bio-fuels (Tanger et al., 2013). Combustion is widely used to produce heat and power from biomass, accounting for over 97% of global bioenergy production, whereas gasification appears to be more suitable for the production of fuels and chemicals from forestry biomass (Sikarwar et al., 2016). For example, gasification of biomass at high temperature (800 – 1100 °C) combined with O₂, steam, and air can produce BioSNG, a mixture of mainly carbon monoxide (CO) and hydrogen (H₂) (Figure 2a). BioSNG can be further processed to produce biofuel through microbial fermentation by *Clostridium* and *Acetobacterium* strains (Munasinghe & Khanal, 2010), or

used directly as building blocks for the production of methanol, methane (CH_4), DME, and other typical end products of gasification (Sikarwar et al., 2017). Recently, heterologous expression of genes involved in the butanol biosynthetic pathway in *Clostridium ljungdahlii* has further improved butanol production (Zhang, Zhao, et al., 2020). Pyrolysis, a thermal degradation process occurring under anoxic conditions, is mainly used to convert dry biomass into gaseous, liquid (bio-oil), and solid (bio-char) intermediates that can be further processed to fuels or chemicals through secondary processes (i.e. hydrogenation, fractionation, gasification, and combustion) (Jahirul et al., 2012). Since dry LCB is well suited to most types of pyrolysis, it can be converted into combined heat and power (CHP), or bio-oils and platform molecules such as 5-hydroxymethylfurfural (5-HMF) and levulinic acid via furfural in a biorefinery process (Melero et al., 2012) (Figure 1). Although liquefaction is suitable for the conversion of biomass feedstock with high water content at low temperatures under high H_2 pressure, it is not as established as pyrolysis owing to expensive costs and complex reactor setup (Gollakota et al., 2018). Torrefaction, a mild form of pyrolysis in an inert reducing environment, is beneficial for the production of homogeneous fuels from biomass with improved physical properties such as grindability and pelletability (Tumuluru et al., 2011). Hydrogenation can be used as an alternative process for the production of renewable diesel, such as hydrotreated vegetable oil from vegetable and animal oils (Dimitriadis et al., 2018).

Anaerobic digestion (AD), fermentation, and transesterification are biochemical processes that can be carried out using biocatalysts for food and agricultural waste management (Cuéllar & Webber, 2008; Xu et al., 2018). In particular, AD is a well-established microbial process comprising three main stages (hydrolysis, acidogenesis, and methanogenesis) for decomposing organic substances under anoxic conditions, resulting in the production of biogas as biofuel and digestate as fertilizer (Table 1 and Figure 2b). Nevertheless, there are several major challenges when applying conventional AD systems to food waste management, mainly due to the heterogeneity of input feedstocks, the type of digester, and various operational factors. To overcome these practical issues, the composition and amount of feedstock need to be checked carefully before entering the digester, which usually requires pretreatments (either single or multiple stages) (Grimberg et al., 2015). In this regard, co-digestion with two or more substrates can be a more effective option in pilot- and commercial-scale AD plants in terms of biofuel yield and process stability. Simultaneous digestion of substrates, when blended at an optimal ratio, can overcome the drawbacks of mono-digestion by diluting harmful compounds, optimizing production, and improving the economic viability of AD plants (Mata-Alvarez et al.,

2014). Moreover, AD with food waste requires appropriate supplementation of micronutrients (e.g., Fe, Ni, Mo, Co, W and Se) and materials (e.g., bio-char) to retain the performance and stability of microbial digesters (Fagbohunge et al., 2017; Romero-Güiza et al., 2016). Recently, a three-stage digester using three separate chambers was examined to prevent fast acidification and inhibition of methanogens when using food waste and horse manure for increased methane yield, based on an assessment of operating costs and system performance (Zhang et al., 2017).

In recent decades, microbial fermentation processes have been employed for industrial production of next-generation biofuels and bio-based chemicals (Devarapalli & Atiyeh, 2015; Green, 2011; Koutinas et al., 2014). Nevertheless, feedstock utilization, microbial platform development, and fermentation process design are poised for further development to achieve sustainable production in competition with synthetic chemicals. Recent advances in biotechnology, genomics, and systems biology enable the design of engineered platform cells for improved production of selected chemicals compared with using native microorganisms (Alper & Stephanopoulos, 2009; Lee et al., 2013; Lee & Lee, 2015). Indeed, many bio-based fuels and a broad range of platform chemicals are now being produced via fermentation routes using engineered microorganisms (Almeida et al., 2012; Lee et al., 2008; Liao et al., 2016; Majidian et al., 2018; Verduyck & De Vos, 2017). In addition to sugar and starch biomass, LCB as agricultural waste can serve as a resource for second generation (2 G) ethanol production using microbial fermentation (Pereira et al., 2015; Saini et al., 2015). Moreover, a variety of microorganisms can also ferment water-miscible or oily substrates to produce 2 G biosurfactants (Banat et al., 2014; Nurfarahin et al., 2018) (Table 2).

LCB provides inexpensive and renewable resources for biotechnological production of biofuels and platform chemicals. However, biomass recalcitrance limits the use of biomass as substrates because it must first be hydrolyzed into fermentable sugars via mechanical or chemical pretreatment before thermochemical or biochemical conversion. To simplify the complex conversion process and minimize costs, CBP has been designed to directly convert LCB into target bio-based products (Brethauer & Studer, 2014; Gowen & Fong, 2011). Unlike conventional biofuel production in which all steps are performed separately, CBP combines enzyme production, polysaccharide hydrolysis, and sugar fermentation into a single-stage operation using single or multiple microorganisms (Figure 2b) (Schuster & Chinn, 2013). A representative example of the implementation of CBP is ethanol production from cellulosic biomass using *S. cerevisiae* expressing cellulase, but the concept of CBP could be applied for the valorization of other types of biomass (Olson et al., 2012). Despite the use of CBP in

industrial processes, quantities of bio-based products from native microorganisms are insufficient to compete with petroleum counterparts. For this purpose, there are three crucial areas in which microorganisms must perform adequately to be effective: product yield to minimize the effects of feedstocks, titer to lower separation costs and decrease the required reactor volumes, and productivity to achieve a rapid rate of production. Thus, microbial platforms for biofuel production should possess the processing capacities for high substrate and high metabolic fluxes, high tolerance to inhibitors and products, and rapid and selective pathways for sugar transport. Advances in metabolic engineering, systems biology, and synthetic biology approaches have facilitated bioprospecting of LCB for high-yield bio-based products using genetically-engineered microorganisms (Gowen & Fong, 2011; Nakamura & Whited, 2003). For instance, *Clostridium cellulovorans* and *Clostridium thermocellum* are promising CBP hosts because they can ferment cellulose; engineered pathways in these microorganisms have been successfully used to divert metabolic flux for the enhanced production of butanol and ethanol (Tian et al., 2019; Yang, Xu, et al., 2015). To this end, the use of microbial consortia has been intensively investigated to develop improved CBP technologies for the production of fuels and chemicals from pretreated LCB. Indeed, CBP incorporating aerobic fungi and anaerobic yeast or bacteria has improved the production yields of lactic acid and ethanol by facilitating the utilization of beech wood and wheat straw as biomass (Brethauer & Studer, 2014; Shahab et al., 2018).

2.3. Biorefineries

A biorefinery combines conversion processes, plants, and facilities to produce biofuels, energy (electricity and heat), and platform molecules from biomass (de Jong & Jungmeier, 2015; Ragauskas et al., 2006; Takkellapati et al., 2018). Thus, different biorefinery configurations can produce a variety of platform intermediates, which is reminiscent of the petrochemical industry (Figure 3). There are several commercially-available biorefinery processes for the production of platform molecules, allowing the use of a single feedstock (either a sugar, starch, or vegetable oil) to produce two or three products, including energy-driven products (Becker et al., 2015; Cok et al., 2014; Luque et al., 2008). These can be directly integrated with existing plants for power, digestion, or biofuel production by providing an appropriate feedstock, except for pretreatment processes for raw materials.

Primary processing (fractionation and hydrolysis) with biomass has been integrated with chemical manufacturing through diversification of product streams involving starch-containing agro-biomass to generate chemical intermediates such as lactic acid (LA), succinic acid (SA), and polyols

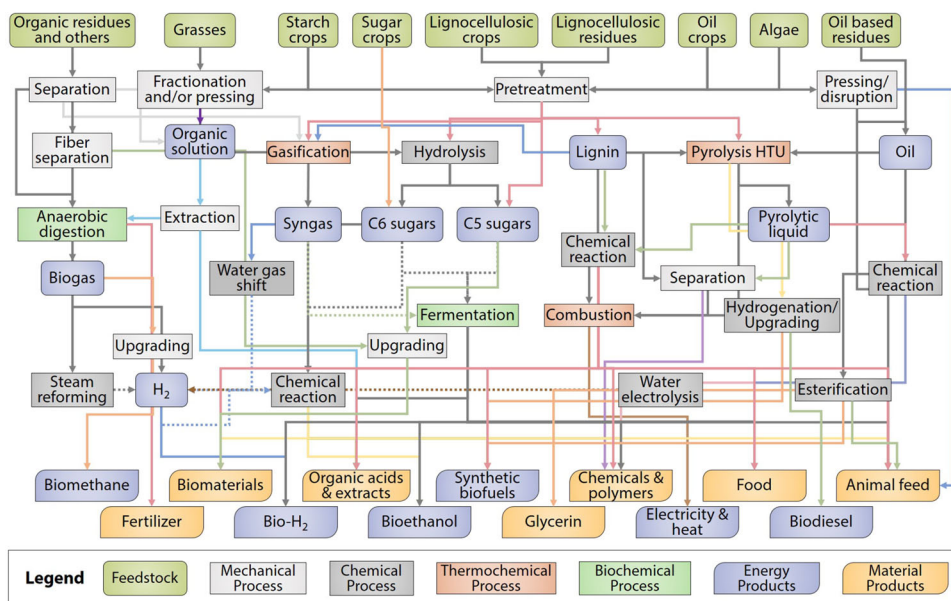


Figure 3. Integrated biorefinery concept for production of bio-based products from biomass feedstock (adapted from (Cherubini et al., 2009)). Mechanical, chemical, and biological processes that valorize inexpensive raw materials to produce bioenergy, biomaterials, and bio-based foods are shown. Individual biorefinery system is distinguished by different colors of lines, and dotted lines indicate links among the biorefinery pathways.

(Figure 1 and see Section 3.1.) (Mussatto & Dragone, 2016; Sauer et al., 2008). Additionally, establishing new biorefinery configurations combined with different feedstocks, platforms, processes, and products has allowed us to utilize lignin, which has great potential for industrial processing. The one-pot reaction strategy integrates biomass fractionation and polymer conversion in a single step, yielding digestible pulp, valorizable lignin, and high-value chemicals (Zang et al., 2020). Moreover, integrated biomass fractionation processes such as the Organocat process maximizes biomass utilization by incorporating catalytic depolymerization in a biphasic solvent system to disentangle the polymer chain structure of LCB and liberate individual components in a single step (Viell et al., 2013; Vom Stein et al., 2011). These advances can improve the value of lignin, and thereby increase the profitability and productivity of biofuels. For example, hemicellulose and lignin, bioactive compounds, and essential oils extracted from the pulping process, wood/bark, and tree species are the major biorefinery types that have benefited from integration with the processing industry (Hamaguchi et al., 2012; Maity, 2015). Furthermore, there is great potential for efficiently deploying biorefinery-based value chains for the co-production of biofuels and high value chemicals in a sustainable manner (Figure 3).

Specifically, biomass gasification produces BioSNG as a platform in energy-driven biorefineries to produce power, heat, Fischer-Tropsch diesel,

Table 3. Fuel characteristics of selected biomass materials (wt%).

Materials	Moisture content (%H ₂ O)	HHV ^a (MJ/kg)	Ash content (%)	References
Fir	6.5	21	0.8	(McKendry, 2002)
Danish pine	8.0	21.2	1.6	(McKendry, 2002)
Willow	60	20.0	1.6	(McKendry, 2002)
Poplar	45	18.5	2.1	(McKendry, 2002)
Cereal straw	6	17.3	4.3	(McKendry, 2002)
Miscanthus	11.5	18.5	2.8	(McKendry, 2002)
Bagasse	45-50	19.4	3.5	(McKendry, 2002)
Switchgrass	13-15	17.4	4.5	(McKendry, 2002)
Bituminous coal	8-12	26-2	8	(McKendry, 2002)
Reed	7.7	17.4	NR ^b	(Kim & Han, 2016)
Larch	10.6	20.1	NR	(Kim & Han, 2016)
Reed canary pellet	<12	>14.5	NR	(Kim & Han, 2016)
Swine manure	12.8	19.5	18.5	(Ro et al., 2010)
Swine manure (Biochar)	3.4	18.3	44.7	(Ro et al., 2010)
Chicken litter	10.2	13.0	30.6	(Ro et al., 2010)
Chicken litter (Biochar)	4.0	13.5	53.2	(Ro et al., 2010)
Rye grass	6.2	18.6	3.2	(Ro et al., 2010)
Corn straw pellet (CSP)	8.79	15.45	9.14	(Xing et al., 2018)
CSP400 (Biochar)	4.67	21.86	22.34	(Xing et al., 2018)
CSP450 (Biochar)	4.84	23.76	24.39	(Xing et al., 2018)
CSP500 (Biochar)	4.53	24.15	25.51	(Xing et al., 2018)
CSP550 (Biochar)	4.42	24.49	26.62	(Xing et al., 2018)
CSP600 (Biochar)	4.46	24.55	27.56	(Xing et al., 2018)
Pine pellets	NR	18.484	NR	(Santos et al., 2016)
Pine pellets (Biochar) 570 1 h	NR	31.747	NR	(Santos et al., 2016)
Pine pellets (Biochar) 570 0.5 h	NR	31.662	NR	(Santos et al., 2016)
Pine pellets (Biochar) 280 1 h	NR	32.253	NR	(Santos et al., 2016)
Pine pellets (Biochar) 280 0.5 h	NR	33.054	NR	(Santos et al., 2016)
Pine pellets (Biochar) 200 1 h	NR	29.735	NR	(Santos et al., 2016)
Pine pellets (Biochar) 200 0.5 h	NR	30.219	NR	(Santos et al., 2016)

^aDry basis, unless stated otherwise.^bNR, Data not reported.

Goldemberg, 2007; Jong et al., 2010; Wang, Dewil, et al., 2019). Recent advances in processing technologies with biomass as feedstocks provide a variety of conversion strategies for producing numerous biofuels and bio-based products (Tables 2 and 3). Furthermore, some have achieved commercially viable production and now compete with their fossil-based counterparts in global markets (Isikgor & Becer, 2015; Jong et al., 2010; Rosales-Calderon & Arantes, 2019). Global biofuel production from agricultural and forestry biomass and biodegradable waste reached 143 billion liters in 2017 (WBA, 2018), and global bioplastics production is expected to increase to 2.6 million tons in 2023 (Díez-Pascual, 2019). In particular, bio-based food and feed ingredients are notable due to numerous potential applications in the food industry (Teekens et al., 2016).

3.1. Bio-based chemicals and building blocks

Chemicals derived from biological conversion of biomass feedstocks can be used as basic and industrial chemicals for a variety of high value-added products (Figure 4). Currently, the market for bio-based building blocks is

dominated by bioethanol (\$58 billion a year), followed by acetic acid and LA, which also constitute a significant market (Taylor et al., 2015). Other sugar derivatives such as xylitol and sorbitol are also established in the market without petrochemical alternatives (Dutta et al., 2012; Ortiz et al., 2013). In particular, bio-based SA is associated with the most dynamic developing market (Cok et al., 2014).

Lactic acid (LA) has a relatively well-established market as a bio-based chemical, with production in demand by both the food industry and polymer production. Indeed, more than half of LA is estimated to be used in the food industry (Aeschelmann et al., 2015). Conventionally, industrial production of LA was achieved using microbial fermentation of sugar from corn, wheat, and cane, alongside chemical synthesis from coal- or crude oil-derived acetaldehyde (Choi et al., 2015) (Figure 5). Currently, over 90% of production is carried out via microbial fermentation (e.g., optimized *Lactobacilli*, engineered yeast, and immobilized *Rhizopus oryzae*), hence using inexpensive biomass as substrates will likely assist the development of cost-effective LA production methods (Hu et al., 2015; Saravanamurugan et al., 2020; Yamane & Tanaka, 2013) (Table 2). Recently, the use of raw materials including starch, lignocellulose, whey, and sugar molasses in microbial fermentation for LA production has been attempted (Hamaguchi et al., 2012; Komesu et al., 2018). In particular, some studies reported LA production titers as high as 99 g/L and 185 g/L via fermentation of non-detoxified wood hydrolysate and sugarcane bagasse hydrolysate with cottonseed meal by *Lactobacillus paracasei* (Kuo et al., 2015) and *Bacillus* sp. P38 (Peng et al., 2014), respectively. Remarkably, after disrupting the D-lactate dehydrogenase gene, *L. paracasei* exhibited high tolerance to inhibitors and achieved production of optically pure LA (Kuo et al., 2015) with high productivity, demonstrating the potential for using agro-industrial wastes and LCB as substrates in large-scale LA production.

Bio-based succinic acid (SA) is one of the fastest growing markets, because it is more price competitive (\$2,940/tons) than petroleum-based SAs (\$2,500/tons) at larger production scales (Taylor et al., 2015). Moreover, it is a very versatile building block that can be transformed into a variety of bio-based chemicals and polymers including 1,4-butanediol (BDO), tetrahydrofuran, and γ -butyrolactone via microbial fermentation integrated with downstream processes (e.g., direct crystallization, electrodialysis, and ammonium sulfate treatment) (Becker et al., 2015; Cok et al., 2014) (Figures 1 and 5). Several companies (e.g., BioAmber, Succinity, Myriant, and Reverdia) are key manufacturers in the SA market (Becker et al., 2015). *Actinobacillus succinogenes*, *Anaerobiospirillum succiniciproducens*, and *Mannheimia succiniciproducens* are employed as key producers in SA production due to their high productivity (Table 2). SA is produced by

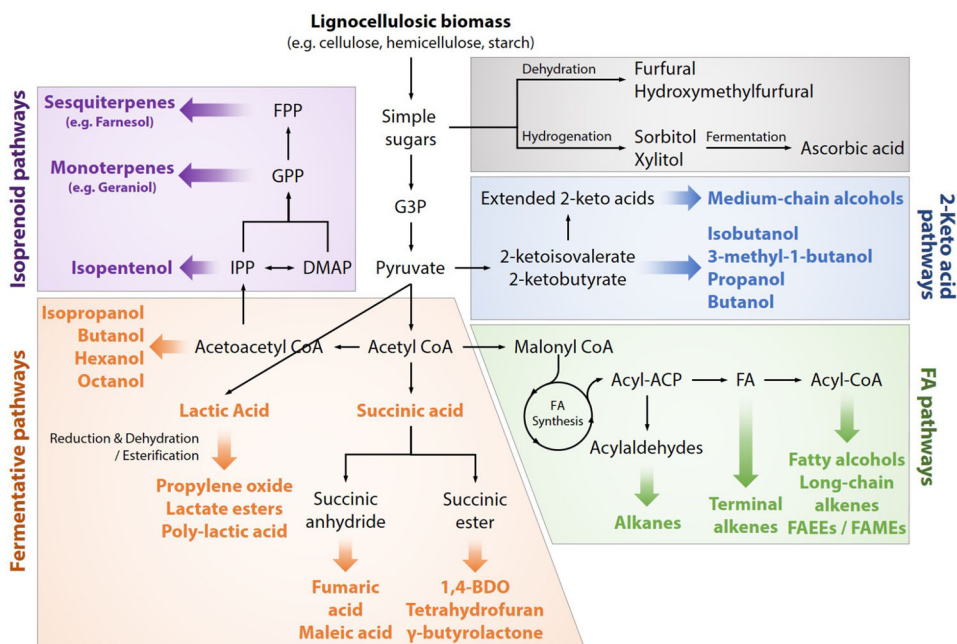


Figure 5. Engineering of metabolic pathways for the production of advanced biofuels and value-added products. The four major targeted pathways in microorganisms that can be manipulated to increase the production of bio-based products are isoprenoid pathways, fermentative pathways, 2-keto acid pathways, and fatty acid pathways, which are colored purple, orange, blue, and green, respectively. The final products from each pathway, including bio-based building blocks, biofuels, bioplastics, and bio-based chemicals, are colored accordingly. 1,4-BDO, 1,4-Butanediol; DMAP, Dimethyl-allylphosphate; FA, Fatty acids; FAE, Fatty acid methyl ester; FAME, Fatty acid ethyl ester; FPP, Farnesyl-pyrophosphate; G3P, Glyceraldehyde-3-phosphate; GPP, Geranyl-diphosphate; IPP, Isopentenyl-diphosphate.

A. succinogenes via anaerobic fermentation of crop stalk hydrolysates, wheat flour, corn stover hydrolysate, and Jerusalem artichoke tuber hydrolysates as substrates. Likewise, *A. succiniciproducens* can produce 47.3 g/L of SA via anaerobic fermentation of waste bread (Leung et al., 2012). However, because they are sensitive to pH and other atmospheric conditions, relatively stable recombinant yeast and *E. coli* strains are often used in SA production (Jansen & van Gulik, 2014). For instance, *E. coli* that has been metabolically-engineered to inactivate competing fermentative pathways produces 98.63 g/L and 127.13 g/L of SA via anaerobic fermentation of cassava pulp (Sawisit et al., 2015) and cassava starch (Chen et al., 2014), respectively.

Other sugar-derived platform molecules include sorbitol and xylitol, which are sugar substitutes used in food, nutraceutical, and cosmetic products (Figure 4). These have little effect on blood sugar and insulin and provide fewer calories (de Kalbermatten et al., 1980). More importantly, they serve as building blocks for other products such as biofuels and commodity

chemicals. Production of sugar-derived molecules involves hydrogenation of D-xylose or glucose or conversion of sugars by microorganisms (Dasgupta et al., 2017; de Albuquerque et al., 2014; Marques et al., 2016). Recent research on microbial conversion of sugars from wastes in food and agricultural industries into sorbitol and xylitol revealed that *Zymomonas mobilis* ATCC 29191 can produce 13.87 g/L of sorbitol through batch fermentation of sugar cane molasses (Cazetta et al., 2005) (Table 2). Moreover, *Candida athensensis* SB18 and *Candida tropiocaalis* produced high concentrations of xylitol from horticultural waste (Zhang et al., 2012) and corncob (Li et al., 2012), respectively. Recently, co-production of ethanol and xylitol by *S. cerevisiae* and *Candida* strains enhanced yields of both products compared to their individual productions (Unrean & Ketsub, 2018; Zahed et al., 2016).

Cellulose-derived 5-Hydroxymethylfurfural (5-HMF) and levoglucosenone (LGO) are other key chemical precursors, with numerous industrial applications in pharmaceutical, plastic, and fuel industries (He et al., 2017). Current production of 5-HMF largely depends on dehydration of C₆ sugars from biomass in a biphasic solvent system (one-pot synthesis) containing acid catalysts (Menegazzo et al., 2018; Teong et al., 2014). For maximum extraction of 5-HMF, the solvents and catalysts, as well as the temperature and time, are important considerations. For instance, 5-HMF was produced with a yield of up to 48% (based on cellulose content), without interference from other biomass components, when corn stover was treated with *N,N*-dimethylacetamide (DMA) containing lithium chloride (LiCl) (Binder & Raines, 2009). Under different conditions, the yield of 5-HMF yield was 57.9% when wheat straw was treated with 1-butyl-3-methylimidazolium chloride containing CrCl₃/LiCl as a catalyst (Wang et al., 2011). Levoglucosenone, an isomer of 5-HMF, can also be produced by pyrolytic recycling of biomass C₆ sugars. LGO yields reached 8.2 wt% from poplar wood and 14.3 wt% from pine when their extraction was catalyzed by solid phosphoric acid and Ni-P-MCM-41 (Li et al., 2020; Zhang et al., 2015). Despite the considerable amount of effort that has been expended in achieving optimal biosynthesis of 5-HMF and LGO, microbial conversion processes still cannot rival the levels achieved by other processes.

3.2. Bioenergy and biofuels

Biomass generated from agriculture, forestry, and ocean industrial waste resources can be transformed through biological and physicochemical conversions into useful forms of energy such as heat, electricity, and transportation fuel (Table 2). In 2017, the 28 countries of the EU (EU-28) produced 758 million tons of oil equivalent (Mtoe) of major energy

production, with renewable energy sources (29.9%) accounting for more than 25% of the EU-28 total production (Eurostat, 2019). Furthermore, the EU-28 reportedly intends to increase the share of renewable energy to reduce energy dependence, with bioenergy as a central contributor. Bioenergy accounts for up to 61% of the total consumption of renewable energy in the EU, and 10% of gross final energy consumption (Calderón et al., 2016). The dependence on bioenergy is increasing nationwide, especially for the heating sector (73%), and basic forms comprised solid biomass fuels, biogas, and transportation biofuels. Bioenergy generates 17.9% of the EU's total renewable power from solid biomass and biogas power plants, and cooperative anaerobic fermentation using agricultural wastes and energy crops is the most common power plant type. The main factors related to the operation of biomass power plants are the continuous supply of low-cost raw materials and CHP. Recently, various biomass CHP systems have been introduced into the market, based on using poultry litter (Jeswani et al., 2019), co-digestion of fruit rind from fluted pumpkin and poultry manure (Dahunsi et al., 2018), and corn stover and distillers grains (Kumar et al., 2010).

Similarly, other countries are developing or applying policies to increase the generation of renewable energy, according to their regional and national policy priorities (i.e., public health, environmental pollution, energy security, and global climate change) (Allen et al., 2016; Fuso Nerini et al., 2018). Interestingly, renewable, hydroelectric, and nuclear energy sectors grew by 17% (equivalent to 69 Mtoe) in 2017, which is above the average for the last 10 years (Dudley, 2018). China has achieved the most growth worldwide with up to 25 Mtoe being generated by renewable, hydroelectric, and nuclear energy sectors.

3.2.1. Solid biomass fuels

Wood biomass is the simplest form of bioenergy that can be easily fueled (Table 3). It can be classified based on size, and both 5 – 100 mm wood chips and wood powder <1 mm are widely available, as are wood pellets (WPs) of a suitable diameter (<25 mm) and briquettes with a much larger diameter (>25 mm) (Stelte, 2012). Wood biomass has a higher energy density than other forms, especially WPs, which are used worldwide as fuels for heating homes and businesses, and for power generation (Bai et al., 2017; Xiao et al., 2015). WPs are manufactured in accordance with the fuel standards set by each country to fulfill fuel requirements according to ISO17225-2 international standards (Masan et al., 2018). In 2016, worldwide production of WPs reached 29 million tons, and imports and consumption of WPs increased by 0.7 million tons in the UK and Korea. Notably, consumption of WPs in Asia increased by 17% (FAO F., 2014),

and WP production using agricultural wastes is expected to reach 50 million tons by 2020 in China (Hu, Yang, et al., 2016). Notably, to increase the energy density of WPs and switch the use of raw materials from wood to various herbaceous plants and agricultural by-products, the scientific community has been working to optimize the quality of WPs by blending wood resources using wood chips and sawdust (Thiffault et al., 2019). Hardwood residues are good sources for generating optimal feedstock blends to prepare pellets with higher density, lower friction, and higher mechanical strength. However, the presence of high concentrations of metals, especially aluminum, was identified as the main issue (Thiffault et al., 2019).

Resources other than wood-based pellets are also candidates for blending with WPs to increase energy density, such as rice straw, barley straw, sunflower stems, chaffs and other agricultural by-products. Some herbaceous plants such as switch grass, miscanthus, and reed canary are good replacements for wood due to their high biomass production. Pellets from these raw materials are known as biomass pellets or agropellets. In Table 3, high heating values (HHVs, in MJ/kg) are compared. Wood-based raw materials possess HHVs of 20 or higher, while values for herbaceous plants and agropellets range from 14 to 20 (Kim & Han, 2016; McKendry, 2002; Ro et al., 2010; Santos et al., 2016; Xing et al., 2018).

Biochar is an interesting biomass because it has higher HHVs than original raw materials (Table 3). After pyrolysis, the HHVs of biochars from corn straw are increased from 15.45 to >20 at typical treatment temperatures (Xing et al., 2018), similar to pine biochars (Santos et al., 2016). In addition, hydrothermal carbonization (HTC) has recently been introduced as a pretreatment process for making an energy-dense solid fuel (biochar) from inefficient LCB (Reza et al., 2012). During this pretreatment, biomass is incubated under hot compressed water at 200 – 260 °C. Pellets obtained from HTC pretreatment possess superior fuel properties than pine pellets and dry torrefied pine pellets, with increased energy density. HTC-pretreated pine pellets at 260 °C can achieve an HHV of 26.4 MJ/kg (Reza et al., 2012). One study compared HTC pretreatment and low-temperature pyrolysis (LTP) for preparing harmonized and carbon-rich biochars (Wang, Qiu, et al., 2019). This study examined the feasibility of HTC and LTP using *Eucommia ulmoides*. Both LTP and HTC treatments contributed to increase the fuel properties of the wood. However, further results confirmed that HTC achieved better grindability, increased hydrophobicity, and lower inorganic content than the LTP process under the same temperature conditions.

Unlike biochars, torrefaction causes biomass to lose more H₂ and O₂ than carbon, and torrefied biomass has lower HHVs (Uslu et al., 2008). To

identify suitable conditions for wet torrefaction treatment, three different types of biomasses were tested from *Humulus lupulus* (HL), *Plumeria alba* (PA), and *Calophyllum inophyllum* (CIL). After wet torrefaction at 260 °C, lower HHVs were obtained for solid fuel torrefied from CIL compared with those from HL and PA, indicating that solid fuels obtained by torrefaction might have the potential to be used as biofuels (Yang, Shimanouchi, et al., 2015). Because WPs are suitable for small and large heating systems, fuel pellets can be made from many different forms of biomass and its residues, as mentioned above. These are all acceptable as biomass fuel pellets, but more research is needed due to their different combustion characteristics.

3.2.2. Biogas

At present, biogas is produced from biomass by AD, and the process is well-established commercially. It can also be generated by thermal gasification of biomass (BioSNG), but this technology is less established. Biogas varies in composition depending on the raw materials used, but it is usually a mixture of trace elements, including CH₄ (50 – 60%), CO₂ (40 – 50%), N₂ (5%), hydrogen sulfide (<1%), and others (Amini et al., 2012; Amini & Reinhart, 2011). There are three different types of biogas; landfill biogas, sewage sludge biogas, and organic waste biogas. Landfill biogas is mainly produced by anaerobic fermentation using organic matter in landfill sites, and first-order equations apply to the generation and collection of landfill biogas. Nevertheless, the collection rate for landfill biogas calculated by modeling was lower than the actual collection rate due to problems in the model parameters used for quantification, including methane generation potential (Amini et al., 2012). The calorific power obtained from landfill biogas is presumably sufficient to power an internal combustion engine (ICE) and generate electricity (Gewald et al., 2012). However, more than 50% of landfill biogas may not be captured properly (Gewald et al., 2012). In addition, landfill biogas is used to replace diesel for vehicle transportation, as exemplified by a biogas plant in Norway that is capable of producing 14,000 m³ of biogas fuel from 50,000 tons of food waste per year, enough to operate 135 municipal buses (Wärtsilä, 2014).

Sewage sludge from the wastewater treatment process is expected to increase in quantity due to an increase in the number of households connected to central wastewater treatment plants (WWTPs), improved efficiency of wastewater treatment, and tightened regulations on wastewater discharge in most countries (Appels et al., 2008). At present, WWTP is used to treat sewage sludge via AD, but interest is growing in optimal biogas production using co-digestion technology after adding co-substrates to sewage sludge. Remarkably, co-digestion of 95% sewage sludge with 5%

crude glycerol resulted in an 8-fold increase in biogas production rate compared with 100% sewage sludge (Maragkaki et al., 2018). Thus, large amounts of fossil fuel could be saved by using alternative biogas produced via sewage sludge fermentation, landfill, and organic waste fermentation (Schneider et al., 2012). Biogas can be burned primarily for heat and power, or converted to CH_4 to generate power for district heating. In one study, 88 sewage treatment plants in Spain were used to estimate capital expenditure for providing district heating, and the estimate was a reduction in CO_2 emissions equivalent to 2,854 M euros (Picardo et al., 2019).

Biogas is also ideal for generating power using CHP systems. The power produced can be publicly supplied or independently used for industrial, agricultural, commercial, or rural settlements (Lantz, 2012). Interestingly, this study concluded that biogas production is not always cost-effective, even when large-scale plants are employed instead of individual farm-scale plants. Rather, economic feasibility can be increased by efficient utilization of reactor volume (Lantz, 2012).

3.2.3. Biofuels

Liquid biofuels for transportation can be categorized into three sub-groups according to substrate type; 1 G biofuels obtained from sugar or starch crops, 2 G biofuels from lignocellulosic residues, and third-generation (3 G) biofuels from aquatic nutritional algae (Table 2) (Favaro et al., 2019; Robak & Balcerak, 2018). The most popular alternative fuel is 1 G bioethanol from yeast fermentation. However, beyond comparison with fossil fuels, production of 1 G biofuels (e.g., biodiesel and bioethanol) in China and Brazil is likely to negatively affect global biodiversity due to habitat loss caused by corn and sugarcane cultivation (Elshout et al., 2019). Hence, other renewable resources such as LCB have been used to produce 2 G biofuels using microbial fermentation integrated with pretreatment processes or CBP (see Section 2.2.). However, numerous technical hurdles must be tackled, including biomass recalcitrance and economic cost issues. During the past two decades, many biomass conversion technologies including biorefining have been introduced to conduct multiple steps in efficient and robust ways using engineered microbial cells or consortiums based on an understanding of cellular and metabolic processes (Brethauer & Studer, 2014; Gowen & Fong, 2011; Himmel et al., 2007; Liao et al., 2016). Alternatively, production of 2 G bioethanol is sometimes conducted in conjunction with the production of biogas. Lignocellulose from corn straw can be converted into bioethanol by yeast fermentation, and the postfermentation residues are then transformed into biogas through CH_4 fermentation (Kotarska et al., 2019). Moreover, yeast fermentation of pretreated cassava peels resulted in combined production of ethanol and CH_4 , with 3-4-fold, and

1.2-1.3-fold increase in energy yield compared to individual production. This type of combined fermentation not only affords the production of two biofuels without waste, but also achieves a higher yield and productivity than conventional ethanol production due to the supply of biodegradable substrates to biomethane-producing bacteria (Moshi et al., 2015).

Advanced biofuel production has been aided by metabolic engineering of microbial pathways in bacteria, yeast, and green algae (Liao et al., 2016; Zhang et al., 2011). The four major pathways targeted to date are (1) fermentative pathways, (2) 2-keto-acid pathways for short- and medium-chain alcohols, (3) isoprenoid pathways, and (4) fatty acid pathways (Figure 5). In the case of *S. cerevisiae*, expression of glycerol utilization genes, heterologous overexpression of key enzymes for fatty acid ester biosynthesis (wax ester synthase/acyl-coenzyme A:diacylglycerol acyltransferase), and deletion of genes involved in glycerol production enhanced the production of fatty acid ethyl esters (FAEEs) from glycerol, with a high FAEE yield and productivity (Yu et al., 2012). Moreover, heterologous expression of a 2-keto acid decarboxylase and an alcohol dehydrogenase (ADH) enabled *E. coli* cells to produce various short-chain alcohols (isobutanol, 3-methyl-1-butanol, 1-propanol, 1-butanol, and others) by diverting flux into amino acid bio-synthetic pathways (Atsumi et al., 2008). Similarly, an engineered *Clostridium cellulolyticum* strain could directly convert crystalline cellulose to isobutanol using its indigenous cellulases and amino acid biosynthesis pathway (Gaida et al., 2016; Higashide et al., 2011). Upon addition of ADH genes from *E. coli* and *Lactococcus lactis*, the pathway from 2-keto acid intermediates was diverted toward alcohol synthesis (Higashide et al., 2011). Using the same approach, a genetically-engineered *Synechococcus elongatus* strain could directly convert CO₂ to isobutanol through photosynthesis, resulting in increased productivity due to overexpression of ribulose-1,5-bisphosphate carboxylase/oxygenase (Atsumi et al., 2009).

Algae, a 3 G feedstock employed widely for biofuel production, has various advantages including increasing O₂ production, replacing human food, minimizing habitat loss, and a low lignin content (Ruiz et al., 2013; Sirajunnisa & Surendhiran, 2016). In addition to these advantages, the US Department of Energy reported a 2- to 5-fold increase in bioethanol production from microalgal fermentation compared with 1 G bioethanol fermentation (Roesijadi et al., 2010; Wargacki et al., 2012). Moreover, recent advances in genome sequencing and editing technologies have enabled the exploitation of a variety of microorganisms, including extremophiles, for the production of new 3 G biofuels such as biodiesel, bioethanol, biogas, biobutanol, and bio-H₂, highlighting the future role of various microbial sources in biofuel and bioenergy production (Chandrasekhar et al., 2015; Lee et al., 2013; Lee & Lee, 2015).

Other essential biofuels from microbial conversion include 1,3-propanediol (1,3-PDO), 2,3-BDO, and butanol (Table 2 and Figure 4). *Citrobacter freundii* FMCC-B 294, under fed-batch fermentation conditions, produced 1,3-PDO using raw glycerol as substrate (Metsoviti et al., 2013). Similar results were observed using *Klebsiella pneumoniae* DSM 4799 under fed-batch conditions with raw glycerol as substrate (Jun et al., 2010). Furthermore, large-scale approaches can be successfully performed using *Clostridium butyricum* AKR102a, since 1,3-PDO was produced in a 200 L fermentation vessel (Wilkens et al., 2012). Additionally, raw glycerol-derived biodiesel is assimilated by *C. butyricum* NRLL B-23495 to produce 1,3-PDO through batch fermentation under anaerobic conditions, while *Enterobacter aerogenes* produces 2,3-BDO in shake-flasks, and *C. freundii* produces ethanol (Metsoviti et al., 2012). Strains of *Klebsiella*, *Enterobacter*, *Serratia* and *Bacillus* are major producers of 2,3-BDO (Song et al., 2019). Three enzymes, acetolactate synthase (Als), acetolactate decarboxylase (AldC), and butanediol dehydrogenase (Bdh), are necessary to convert pyruvate to α -acetolactate, acetoin, and 2,3-butanediol, respectively (Yang et al., 2016). For instance, *Enterobacter cloacae* produces 93.9 g/L of 2,3-BDO from cassava powder in fed-batch SSF (Wang et al., 2012). To improve the efficiency of 2,3-BDO production, genes involved in byproduct-producing paths were knocked out. For instance, deletion of lactate dehydrogenase diverted carbon flux from lactate production to 2,3-BDO biosynthesis in *E. aerogenes* (Jung et al., 2012).

3.3. Bioplastics/biomaterials

Bioplastics are generally comprised of polyhydroxyalkanoate (PHA) and other biopolymers such as polylactide (PLA), polybutylene succinate (PBS), and bio-polyethylene terephthalic acid (bio-PET), and these biodegradable plastics are biosynthesized by various microorganisms including *Caulobacter segnis* (Bustamante et al., 2019), *Ralstonia eutropha* (Yu & Si, 2004), *Pseudomonas putida* (Możejko-Ciesielska & Serafim, 2019), *Yarrowia lipolytica* (Rigouin et al., 2019), *Xanthomonas oryzae* (Long et al., 2018), and others. Bioplastics are classified into three major classes based on their monomer production routes. The first class includes naturally-derived biomass polymers composed of direct-use biomass obtained by chemical modification of cellulose, cellulose acetate, starches, chitin, and other compounds (Tsang et al., 2019). The second class comprises biopolymers synthesized using industrial or engineered microorganisms to produce PHAs and similar molecules. The third class of bioplastics are bio-synthetic polymers comprising PLA, PBS, and bio-PET (Kimura, 2009; Salvador et al., 2019). Bio-polyethylene, bio-PET, and polybutylene terephthalic acid have

already been, or soon will be, introduced to the commercial market, and polyether-ether ketone and bio-based liquid crystalline polymers have been thoroughly studied at the laboratory scale (Nakajima et al., 2017). As mentioned above, bioplastics are commercially available and widely used in environmentally friendly, flexible packaging applications (Papadopoulou et al., 2019; Zhang, Xiao, et al., 2020), as well as agricultural and medical products (Rujnić-Sokele & Pilipović, 2017). To produce various bio-based monomeric building-block chemicals from renewable feedstocks, metabolic engineering tools and strategies were employed to generate biomonomer-producing engineered microbial platform cells (Adkins et al., 2012). For the production of bio-based polyamide (PA) used in the textile industry, there have been several attempts to produce biomonomers (e.g., putrescine and cadaverine) as new classes of renewable bioplastics for polyamide homopolymers. Overexpression of ornithine decarboxylase rendered a metabolically engineered *E. coli* strain able to produce putrescine at high titer (Qian et al., 2009). Similarly, *Corynebacterium glutamicum* engineered by genomic insertion of lysine decarboxylase achieved enhanced production of cadaverine (Mimitsuka et al., 2007). Additional insertion of an α -amylase gene into *C. glutamicum*, resulted in the production of 23.4 mM cadaverine from soluble starch (Tateno et al., 2009).

The marketability of lignocellulosic polymers is growing rapidly due to impressive stability and biocompatibility, and cellulose acetate (CA) is the world's most abundant biopolymer (Mallepally et al., 2015). CA is used for food and medical purposes, and much research has focused on the generation of CA structures, including some bioactive compounds (Milovanovic et al., 2015; 2016). Meanwhile, lignin in the form of lignin sulfonate is applied to a variety of low-cost products, mainly as a dispersant in cement admixtures and dye solutions. For optimal use, lignin sulfonates should be of an appropriate molecular weight and possess a suitable degree of sulfonation (Aro & Fatehi, 2017). Approximately 2.1 million tons of bioplastics were produced globally in 2019, and packaging remains the main application, accounting for more than 53% (1.14 million tons) of the total bioplastics market in 2019 (nova-Institute, 2019).

3.4. Bio-based food and feed ingredients

Bio-based food and feed ingredients have numerous applications in various sectors as dairy products, beverages, animal feeds, and bakery goods. Hence, they constitute a growing market with a huge demand. In the case of amino acids, to meet the increased industrial demand for millions of ton per year for food and feed additives, various engineered microorganisms have been exploited to improve the production yields by optimization of

biosynthetic pathways and carbon flux in central metabolism (Bongaerts et al., 2001). Currently, representative manufacturers (e.g., Ajinomoto, Amino, Cargill, CJ cheiljedang, and Evonik) are producing food-grade amino acids through microbial fermentation of cheap substrates (Cho et al., 2013; Kirchner & Tauch, 2003). One of the major work-horse strain is *C. glutamicum*, which produces L-glutamate, L-lysine, and L-valine from sugars, organic acids, and alcohols. Remarkably, metabolic engineering of *C. glutamicum* not only enhanced the productivity but also broadened the substrate spectrum to non-utilizable biomass and wastes such as rice straw and wheat bran hydrolysates (Gopinath et al., 2011; Meiswinkel et al., 2013), crude glycerol (Meiswinkel et al., 2013; Rittmann et al., 2008), soluble starch (Seibold et al., 2006) and silage juices (Neuner et al., 2013). Recently, clustered regularly interspaced short palindromic repeats interference (CRISPRi)/dCas9-based engineering of *C. glutamicum* facilitated metabolic pathway remodeling for the hyperproduction of target products (Cleto et al., 2016). Analogously, engineered *E. coli* strains have been extensively exploited for the production of amino acids (Buchholz et al., 2013), and engineered *Pantoea ananatis* is also a potential host for the production of cysteine due to its natural resistance to low pH and saturated amino acid concentrations (Takumi et al., 2017).

Bioactive peptides are functional ingredients that exert beneficial effects on human physiology, beyond their conventional nutritional values. Bioactive peptides can act as antioxidants, antimicrobial agents, anti-inflammatory drugs, antihypertensive compounds that inhibit angiotensin-converting enzyme (ACE), and cytomodulatory effectors (Kang et al., 2020). Industrial waste is often protein-rich and may therefore provide an alternative for preparing bioactive compounds. For example, bioactive peptides can be obtained from proteinaceous wastes such as soybean meal, residues from olive production, rapeseed meal, chicken feathers, fish waste, and egg protein (Lemes et al., 2016). In particular, marine organisms are abundant resources of nitrogenous components. Fish waste components contain 10–23% (w/w) protein with potential for mining of biofunctional peptides (Harnedy & FitzGerald, 2012). In addition to marine-derived bioactive peptides, food and agricultural feedstocks are also used as substrates. For instance, crude goat whey, a major food waste, is also used as a substrate to produce ACE inhibitory peptides through aerobic fermentation by *Kluyveromyces marxianus* and *Lactobacillus rhamnosus* (Hamme et al., 2009). Moreover, low-molecular-weight skin anti-aging peptides were produced from feather hydrolysate obtained by AD of poultry feathers using the extremophilic bacterium *Fervidobacterium islandicum* AW-1 (Jin et al., 2018; 2019; Yeo et al., 2018).

Many sugar-based biopolymers, such as xanthan gum and chitin, are widely used as food ingredients due to rheological features that make them

suitable for viscosifying and stabilizing food products over a wide range of temperatures and pH (Palaniraj & Jayaraman, 2011). Currently, NovaXan (Archer-Daniels-Midland), GRINDSTED (DuPont), and Satiaxane (Cargill) are producing bacteria-derived polysaccharides derived from biomass through microbial fermentation with *Xanthomonas campestris* (Figures 1 and 4). For example, under optimized fermentation conditions, *X. campestris* can use various biomass substrates including cheese whey (a by-product of the dairy industry) (Mesomo et al., 2009), sugar beet molasses (a co-product of sugar production) (Kalogiannis et al., 2003), and molasses with chicken feather peptone (Ozdal & Başaran Kurbanoglu, 2019) to produce xanthan gum, an important industrial biopolymer (Table 2). Likewise, dextrans can be produced by an aerobic culture of *Leuconostoc mesenteroides* from sugar beet molasses (Vedyashkina et al., 2005) and carob pod extract (Santos et al., 2005). In addition, gellan, used as a thickening and gelling agent in many food applications, can be produced commercially via microbial fermentation using *Sphingomonas paucimobilis* (Fialho et al., 2008).

Chitin is a natural polysaccharide found in crustaceans, mollusks, insects, and fungi. Processing of seafood results in exoskeleton and cephalothorax waste materials, which contain valuable ingredients including chitin. Conventionally, recovery of chitin from these wastes involves a two-step chemical procedure of demineralization and deproteinization using strong acids and bases (Arbia et al., 2013; Younes & Rinaudo, 2015). Such procedures also suffer from significant disadvantages: depolymerization may affect the innate properties of polymers, and energy consumption may threaten the environment. Various biological processes have been investigated to replace hazardous chemical processes (Gortari & Hours, 2013). For example, microbial cells including *Lactobacillus* strains provide an attractive method over chemical extraction, which are beneficial for the recovery of chitin from crustacean wastes (Khanafari et al., 2008). In the case of shrimp shell waste, its fermentation by *Bacillus cereus* showed high levels of efficiency (97.1% and 95%) in deproteinization and demineralization, respectively (Sorokulova et al., 2009). Co-fermentation of prawn waste by *L. lactis* and *Teredinibacter turnerae* also achieved a high yield for deproteinization (95%) and demineralization (95%) processes (Aytekin & Elibol, 2010) (Table 2).

Carotenoids have been traditionally used as food colorings for many years, and their applications now extend to animal feeds and dietary supplements, owing to their physiological benefits that include anti-carcinogenic, immune modulatory, and antioxidant properties. However, there are serious safety issues associated with artificial synthetic colorants added to food, which has led to worldwide interest in the production of pigments from natural sources (Taskin & Erdal, 2011). Due to increasing demand,

various types of carotenoids, including beta-carotene and astaxanthin, are produced by various yeast strains from low-cost substrates (Table 2). *Rhodotorula glutinis*, a carotenoid-producing yeast, uses raw materials of agro-industrial origin to produce carotenoids. Using sugarcane molasses (Aksu & Eren, 2005; Bhosale & Gadre, 2001) and loquat kernel (Taskin & Erdal, 2011), fed-batch and batch fermentation of *R. glutinis* produces carotenoids. Moreover, astaxanthin can be produced by *Xanthophyllomyces dendrorhous* (*Phaffia rhodozyma*) via fed-batch fermentation using rapeseed meal hydrolysate, a by-product of oil processing (Harith et al., 2019), and mustard waste hydrolysate (Tinoi et al., 2006). Remarkably, *X. dendrorhous* can produce astaxanthin at 100 L-scale fed-batch fermentation using sugar beet molasses, suggesting a possibility of an industrial scale production (An et al., 2001).

Prebiotics, including galacto-oligosaccharides, fructo-oligosaccharides (FOS), lactulose, lactosucrose, and inulin, are non-digestible oligosaccharides that serve as substrates for probiotics. To the host, prebiotics provide various health benefits including immunomodulatory effects, antioxidant properties, and anti-cancer activities. Thus, prebiotic compounds are applied to fortify food products and animal feeds, supplemented in cosmetics and pharmaceuticals, and even added to agricultural products (Varzakas et al., 2018). To date, most processes for preparing food-grade prebiotics from biomass have involved chemical synthesis or enzymatic production using glycosidases or glycosyl transferases (Panesar et al., 2013). However, microbial production of FOS from agro-industrial residues (e.g., corn cobs, coffee silverskin, and cork oak) was successfully demonstrated using *Aspergillus japonicus* and solid-state fermentation (Mussatto & Teixeira, 2010) (Table 2).

3.5. Biosurfactants

Surfactants possess unique properties, including the ability to (1) radically modify interfacial properties, and (2) self-assemble and form micelles. These unique properties facilitate their application in various industries including agricultural chemicals, cosmetics and personal care products, detergents, and foods, because they can improve texture, stabilize emulsions, and modify rheological properties (Figure 4). The largest market for surfactants is household cleaning detergents, followed by personal care products (Schramm et al., 2003). While 1 G surfactants are petroleum-based and chemically synthesized, 2 G biosurfactants can be manufactured from renewable feedstocks through microbial fermentation. In recent years, biosurfactants have received increasing attention as promising alternatives to 1 G surfactants, over which they offer several advantages including lower toxicity, lower critical micelle concentrations, higher ecological

acceptability, better structural diversity, and higher specific activity under extreme conditions (Kitamoto et al., 2002; Makkar et al., 2011). Recently, Evonik (Germany) has commercialized bio-based sophorolipids from *Starmerella bombicola* for use in cleaning solutions (Jezierska et al., 2019). However, low yields, high production cost, and complicated downstream processes for biosurfactant production limit their market penetration (Wisnuprpto et al., 2011). Moreover, the highly branched structures of biosurfactants have been associated with slower or poorer biodegradability than linear analogs. In this regard, many regulatory agencies in Europe and the United States mandate standardized testing for the biodegradability of products used in consumer and industrial applications (Bragin et al., 2020). Croda International PLC (U.K.) have manufactured bio-based ethoxylated surfactants by incorporating bio-based ethylene oxide into bio-based fatty alcohol (Ahuja & Singh 2020; Marchant & Banat, 2012). Nevertheless, to compete economically with chemical surfactants, the use of inexpensive raw materials such as agricultural or food waste is key to reducing the final product cost of biosurfactants (Makkar & Cameotra, 1999).

Various renewable sources and agro-industrial or food wastes have been used for low-molecular weight biosurfactant production using microorganisms (Figure 4). Waste frying oil, molasses, animal fat, industrial waste (crude glycerol), and whey are examples of low-cost substrates that have been employed for microbial growth (Table 2). *Candida bombicola* and *Candida apicola* are major producers of sophorolipids. Microbial fermentation of sweet sorghum bagasse supplemented with yellow grease using *C. bombicola* produced 35.9 g/L of sophorolipids with a high production yield of 0.56 g/g (Samad et al., 2017). Moreover, *Pseudomonas aeruginosa* is a well-known producer of rhamnolipids, key components of microbial biosurfactants. Recently, it was reported that *P. aeruginosa* can produce rhamnolipids from waste frying coconut oil (George & Jayachandran, 2013) and soybean soapstock (Nitschke et al., 2005) as the sole carbon source. In particular, biosurfactant production by *P. aeruginosa* in a 3 L-scale fermenter potentiated industrial production using waste motor lubricant oil and peanut oil cake (Thavasi, Nambaru, et al., 2011). These glycolipids offer a wide range of applications in the pharmaceutical and cosmetic fields. Moreover, low molecular weight surfactants (lipopeptides) are produced by *Bacillus* strains. *Bacillus subtilis* MTCC 2423 produces surfactin, another key component of biosurfactants, from waste frying sunflower and rice bran oils (Vedaraman & Venkatesh, 2011).

3.6. Biolubricants

Biolubricants are mainly produced using vegetable oils such as canola (Sharma et al., 2015), sunflower (Quinchia et al., 2009), soybean, and palm

(Igartua et al., 2009), and can be used in devices that leak oil directly into the environment, as well as in total loss lubrication (Figure 4). Biolubricants are particularly useful for two-stroke engines, chainsaw bars and chains, and marine applications to provide a clean and safe working environment totally free from aromatic pollutants (Igartua et al., 2009; Quinchia et al., 2009). Canola oil and canola biodiesel are promising candidates for producing biolubricants because they possess potent lubricity properties. Canola biodiesel-based biolubricants possess low viscosity, making them ideal for use in automobiles, while canola oil biolubricants are useful for high temperature conditions due to high thermal stability (Sharma et al., 2015). In one recent study, high oleic acid-containing sunflower oil was mixed with ethylene vinyl acetate and styrene-butadiene-styrene at a range of concentrations from 0.5% to 5% (w/w) (Quinchia et al., 2009). These two polymers increased the viscosity of the oil in a concentration-dependent manner, hence blending oils with ethylene vinyl acetate might be suitable for preparing lubrication agents for bearings and four-stroke engines.

4. Barriers to the development of biological routes to bio-based products

4.1. Feedstock-related barriers

Feedstock-related issues of biomass conversion to bio-based products are growing in importance. The most important restraint is the gap between supply and demand. Indeed, large-scale production of 1 G biofuels from sugar or starch crops may threaten food security. Thankfully, the recent development of 2 G and 3 G biofuels from woody biomass and microalgae does not directly compete with food production (Eickhout, 2012). Moreover, the continuous supply of biomass of adequate quality in sufficient quantities at a consistent price throughout the year is the major challenge for biomass feedstocks. Irregularity in supply results in variable feedstock price, which ultimately increases the cost of the final product. For instance, biomass feedstock costs in Asia and Europe are high, partially due to the seasonality of different biomass types. Also, vegetable oils as bio-lubricants have high biodegradability and good lubricity, but exhibit very low thermo-oxidative stability and a narrow range of viscosity, and should therefore be improved to develop suitable biolubricants (Salih et al., 2013). Moreover, expensive biomass feedstocks and bio-based products cannot compete with much cheaper fossil-based equivalents, and marginal land for feedstock production is insufficient due to an increase in biomass utilization. Thus, to produce biomass feedstocks at a competitive cost, vast amounts of arable land must be secured without jeopardizing non-energy

needs (Lewandowski, 2015). Furthermore, excessive pretreatment is required before using MSW as substrates, which inevitably increases the overall production cost.

4.2. Process-related barriers

For bio-based products to compete with fossil-based products, higher production titers and yields are essential. To this end, major hurdles to overcome include (1) avoiding dependence on expensive and complex nutrient sources, (2) engineering heterologous fermentative pathways, and (3) constructing biological systems that tolerate high substrate concentrations. In the case of (1), biomass is a complex heterogeneous mixture of various natural components, which limits substrate availability and decreases the efficiency of the overall production process. Therefore, for efficient biomass utilization, it is necessary to design and incorporate pretreatment and downstream processes to generate a homogeneous substrate (Irmak, 2019). For instance, various pretreatment processes are available for separation of LCB components; physical, physicochemical, and biological pretreatments can be applied to maximize biomass conversion (Maurya et al., 2015). Regarding (2), current bio-based production using microorganisms results in lower product yields than chemical synthesis, due to inhibitory effects of pretreatment processes and inherent limitations in metabolic capabilities (Jönsson & Martín, 2016). The design of microbial cell factories with increased tolerance to inhibitors or that divert metabolic flux is expected to facilitate the efficient conversion of biomass for the production of biofuels and bio-based products. Indeed, engineered *E. coli* and *S. cerevisiae* have been widely used as platform organisms in commercial-scale production due to the wealth of genetic manipulation tools available for these model research organisms. Moreover, recent advances in metabolic engineering, synthetic biology, and genome engineering enable us to manipulate other native microorganisms (Keasling, 2010; Lee et al., 2013; Shapiro et al., 2018). For instance, inhibitor-tolerant and hyper-butanol producing *Clostridium* strains have been reported to promote the production of butanol and acetone-butanol-ethanol (Baral & Shah, 2014). Despite significant progress, there remain countless unexplored native microorganisms that could potentially serve as industrial strains, but work is needed to develop appropriate genetic tools and methods for their use in the production of target products (Coradini et al., 2015). To address (3), systems metabolic engineering approaches can be employed to adapt microbial strains to enhance their tolerance to high substrate concentrations and product titers (Lee & Kim, 2015). Additionally, alongside tolerance to high substrate concentrations, waste streams may be recycled to reuse a portion of substrates

in subsequent production processes (e.g., production of PHA from whey) (Koller, 2015). Indeed, this approach can make a process more sustainable and cost-effective, but this is a relatively unexplored area.

4.3. Industry- and market-related barriers

Despite a predicted high compound annual growth rate (CAGR) of 10.5% by the end of 2023, current bio-based chemicals are considered more of a niche market than their chemically synthesized counterparts (2019). Implementing new technologies can rapidly reduce overall production costs, and this is essential for the commercialization and market penetration of bio-based products. However, in many biorefinery processes, this is not the case, a cost-effective pretreatment may not be available. There are more than 100 algae-to-fuel companies producing bioenergy and biofuels from algal biomass, but a commercial facility has not yet been built. A highly modified enzyme system for the saccharification step and an appropriate microorganism for converting biomass to ethanol are required for successful biofuel production in very large volumes at low cost (Himmel et al., 2007; John et al., 2011). Moreover, production at a large scale is key for commercialization of products, but only a few bio-based products are economically favorable compared with their fossil-based equivalents. In the case of PHAs, the major obstacle to wide acceptance is their high price, which is mainly due to the high cost of polymer recovery. In recent years, considerable effort has been made to improve production yields for various PHAs at pilot plant scale (Du et al., 2004). However, there remain large hurdles preventing full commercial production of PHAs, not least high production cost and low product yield.

Given that expensive biomass feedstocks and bio-based production cannot compete with much cheaper fossil-based equivalents, better performance and/or unique properties of bio-based products should be sought. In general, bio-based products are regarded as more environmentally friendly than petroleum-based products. However, there is still a lack of a holistic approach to examining their environmental profiles, without which no claims can be made about their better sustainability (Walker & Rothman, 2020). To measure and analyze the environmental impact, including GHG, an end-to-end analysis of the product and process should be done using life cycle assessments (LCA) and product carbon footprints. In addition, there is a huge gap between policy and subsidy frameworks with respect to conflicts between sustainability goals and market needs. To partly overcome these issues and boost sustainable bio-based products, both standardization and certification schemes on clear labeling for renewable resources and bio-based products are urgently required. Very recently, analysis of the

LCA of commercialized bio-based chemicals produced via microbial fermentation was carried out to quantify environmental sustainability performance (Ögmundarson et al., 2020). The results suggest that LCA should include a broader range of impact indicators beyond reducing global warming impacts over fossil-based products (Liu & Rajagopal, 2019). Moreover, LCA-aided biotechnologies, including new roles for microbes in bio-based production, should be employed systematically to solve existing sustainability challenges.

5. Conclusions and future perspectives

Herein, we provide an overview of recent progress in biological routes to bio-based products using raw materials, wastes, and residues as renewable resources (Figure 3). We also delineate bio-based products based on bio-mass feedstock, process type, biorefineries, and practical barriers (Table 2). Estimating the potential availability of bio-based products varies significantly. Indeed, there are still several technical and economic barriers to fully exploiting biomass feedstocks for bio-based products and biofuels (Kretschmer et al., 2013). In particular, organic wastes, including agri-food wastes, have great potential to be valorized using various biorefineries combined with down-stream processes for bio-based production. Notably, instead of current microbe-based biofuels derived from oils, carbohydrates, and fats, there have been recent attempts to generate microbes that utilize and/or produce proteins refined to prepare biofuels, which differs from traditional biofuel production routes (Li et al., 2018; Mielenz, 2011). The development of innovative processes together with the valorization of their processing residues will enhance these supply and value chains, making them practically feasible. In the case of bio-based chemicals, lower dependency on petroleum and feedstock diversification can make sustainability more attractive to manufacturers, and result in innovative products. The future of bioenergy is likely to be geared toward energy generation and the production of high value-added chemicals to achieve cost-effective bioenergy production. Such ‘all-in-one’ strategies can be achieved through newly developed biorefineries applicable for a broad range of technical systems in a sustainable way (Figure 1).

Recent advances in biotechnology-aided biorefineries using biocatalysts have made possible a prosperous future bioeconomy. However, the shape of the bioeconomy will be affected by many external factors such as location, the sizes and types of product segmentation, population, income, consumers, energy consumption, the availability of natural resources and their cost, and available technologies. For the future bioeconomy to be realized, targeted research and global cooperation across different fields are required,

as are secure and responsible biomass usage and climate action. Furthermore, to compete with conventional, fossil-derived counterparts, the overall industry should focus on responsible usage and efficient upgrading of biomass as well as integration of valorized process for the production of high value-added products. Collectively, together with the implementation of suitable policies, these approaches could facilitate the future design and development of the bioeconomy, while preserving biodiversity and protecting the environment.

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References

- Adkins, J., Pugh, S., McKenna, R., & Nielsen, D. R. (2012). Engineering microbial chemical factories to produce renewable “biomonomers”. *Frontiers in Microbiology*, 3, 313 <https://doi.org/10.3389/fmicb.2012.00313>
- Aeschelmann, F., Carus, M., Aeschelmann, F., Carus, M., Baltus, W., Blum, H., Busch, R., Carrez, D., Ißbrücker, C., Käß, H., Lange, K.-B., Philp, J., Ravenstijn, J., & von Pogrell, H. (2015). Market study and trend reports on bio-based building blocks and polymers in the world – capacities, production and applications: status quo and trends towards 2020.
- Ahuja, K., Singh, S. (2020). *Biosurfactants market size by product (sophorolipids, rhamnolipids, alkyl polyglucosides [APG], methyl ethyl sulfonates [MES], sucrose esters, sorbitan esters, lipopeptides), by application (household detergents, personal care, industrial cleaners, food processing, oilfield chemicals, agricultural chemicals, textiles), industry analysis report, regional outlook, application potential, price trend, competitive market share & forecast, 2020 – 2026*. <https://www.gminsights.com/methodology/detail/biosurfactants-market-report>
- Aksu, Z., & Eren, A. T. (2005). Carotenoids production by the yeast *Rhodotorula mucilaginosa*: Use of agricultural wastes as a carbon source. *Process Biochemistry*, 40(9), 2985–2991. <https://doi.org/10.1016/j.procbio.2005.01.011>
- Alexandratos, N., & Bruinsma, J. (2012). World agriculture towards 2030/2050: The 2012 revision. FAO.

- Allen, C., Metternicht, G., & Wiedmann, T. (2016). National pathways to the sustainable development goals (SDGs): A comparative review of scenario modelling tools. *Environmental Science & Policy*, 66, 199–207. <https://doi.org/10.1016/j.envsci.2016.09.008>
- Almeida, J. R., Favaro, L. C., & Quirino, B. F. (2012). Biodiesel biorefinery: Opportunities and challenges for microbial production of fuels and chemicals from glycerol waste. *Biotechnology for Biofuels*, 5(1), 48 <https://doi.org/10.1186/1754-6834-5-48>
- Alper, H., & Stephanopoulos, G. (2009). Engineering for biofuels: Exploiting innate microbial capacity or importing biosynthetic potential? *Nature Reviews. Microbiology*, 7(10), 715–723. <https://doi.org/10.1038/nrmicro2186>
- Amini, H. R., & Reinhart, D. R. (2011). Regional prediction of long-term landfill gas to energy potential. *Waste Management (New York, N.Y.)*, 31(9-10), 2020–2026. <https://doi.org/10.1016/j.wasman.2011.05.010>
- Amini, H. R., Reinhart, D. R., & Mackie, K. R. (2012). Determination of first-order landfill gas modeling parameters and uncertainties. *Waste Management (New York, N.Y.)*, 32(2), 305–316. <https://doi.org/10.1016/j.wasman.2011.09.021>
- An, G.-H., Jang, B.-G., & Cho, M.-H. (2001). Cultivation of the carotenoid-hyperproducing mutant 2A2N of the red yeast *Xanthophyllomyces dendrorhous* (*Phaffia rhodozyma*) with molasses. *Journal of Bioscience and Bioengineering*, 92(2), 121–125. [https://doi.org/10.1016/S1389-1723\(01\)80211-5](https://doi.org/10.1016/S1389-1723(01)80211-5)
- Andler, S. M., & Goddard, J. M. (2018). Transforming food waste: How immobilized enzymes can valorize waste streams into revenue streams. *NPJ Science of Food*, 2, 19 <https://doi.org/10.1038/s41538-018-0028-2>
- Appels, L., Baeyens, J., Degreve, J., & Dewil, R. (2008). Principles and potential of the anaerobic digestion of waste-activated sludge. *Progress in Energy and Combustion Science*, 34(6), 755–781. <https://doi.org/10.1016/j.peccs.2008.06.002>
- Arapoglou, D., Varzakas, T., Vlyssides, A., & Israilides, C. (2010). Ethanol production from potato peel waste (PPW). *Waste management (New York, N.Y.)*, 30(10), 1898–1902. <https://doi.org/10.1016/j.wasman.2010.04.017>
- Arbia, W., Arbia, L., Adour, L., & Amrane, A. (2013). Chitin extraction from crustacean shells using biological methods—a review. *Food Technology and Biotechnology*, 51, 12–25.
- Aro, T., & Fatehi, P. (2017). Production and application of lignosulfonates and sulfonated lignin. *Chemosuschem.*, 10(9), 1861–1877. <https://doi.org/10.1002/cssc.201700082>
- Atsumi, S., Hanai, T., & Liao, J. C. (2008). Non-fermentative pathways for synthesis of branched-chain higher alcohols as biofuels. *Nature*, 451(7174), 86–89. <https://doi.org/10.1038/nature06450>
- Atsumi, S., Higashide, W., & Liao, J. C. (2009). Direct photosynthetic recycling of carbon dioxide to isobutyraldehyde. *Nature Biotechnology*, 27(12), 1177–1180. <https://doi.org/10.1038/nbt.1586>
- Aytekin, O., & Elibol, M. (2010). Cocultivation of *Lactococcus lactis* and *Teredinobacter turnirae* for biological chitin extraction from prawn waste. *Bioprocess and Biosystems Engineering*, 33(3), 393–399. <https://doi.org/10.1007/s00449-009-0337-6>
- Bacenetti, J., Restuccia, A., Schillaci, G., & Failla, S. (2017). Biodiesel production from unconventional oilseed crops (*Linum usitatissimum* L. and *Camelina sativa* L.) in Mediterranean conditions: Environmental sustainability assessment. *Renewable Energy*, 112, 444–456. <https://doi.org/10.1016/j.renene.2017.05.044>
- Bai, X., Wang, G., Gong, C., Yu, Y., Liu, W., & Wang, D. (2017). Co-pelletizing characteristics of torrefied wheat straw with peanut shell. *Bioresource Technology*, 233, 373–381. <https://doi.org/10.1016/j.biortech.2017.02.091>

- Banat, I. M., Satpute, S. K., Cameotra, S. S., Patil, R., & Nyayanit, N. V. (2014). Cost effective technologies and renewable substrates for biosurfactants' production. *Frontiers in Microbiology*, 5, 697 <https://doi.org/10.3389/fmicb.2014.00697>
- Baral, N. R., & Shah, A. (2014). Microbial inhibitors: Formation and effects on acetone-butanol-ethanol fermentation of lignocellulosic biomass. *Appl Microbiol Biotechnol*, 98(22), 9151–9172. <https://doi.org/10.1007/s00253-014-6106-8>
- Becker, J., Lange, A., Fabarius, J., & Wittmann, C. (2015). Top value platform chemicals: Bio-based production of organic acids. *Current Opinion in Biotechnology*, 36, 168–175. <https://doi.org/10.1016/j.copbio.2015.08.022>
- Bengtsson, M., Alfredsson, E., Cohen, M., Lorek, S., & Schroeder, P. (2018). Transforming systems of consumption and production for achieving the sustainable development goals: Moving beyond efficiency. *Sustainability Science*, 13(6), 1533–1547. <https://doi.org/10.1007/s11625-018-0582-1>
- Ben-Iwo, J., Manovic, V., & Longhurst, P. (2016). Biomass resources and biofuels potential for the production of transportation fuels in Nigeria. *Renewable and Sustainable Energy Reviews*, 63, 172–192. <https://doi.org/10.1016/j.rser.2016.05.050>
- Bhatia, S. K., Gurav, R., Choi, T.-R., Jung, H.-R., Yang, S.-Y., Moon, Y.-M., Song, H.-S., Jeon, J.-M., Choi, K.-Y., & Yang, Y.-H. (2019). Bioconversion of plant biomass hydrolysate into bioplastic (polyhydroxyalkanoates) using *Ralstonia eutropha* 5119. *Bioresource Technology*, 271, 306–315. <https://doi.org/10.1016/j.biortech.2018.09.122>
- Bhosale, P., & Gadre, R. (2001). beta-Carotene production in sugarcane molasses by a *Rhodotorula glutinis* mutant. *Journal of Industrial Microbiology & Biotechnology*, 26(6), 327–332. <https://doi.org/10.1038/sj.jim.7000138>
- Bhutto, A. W., Qureshi, K., Abro, R., Harijan, K., Zhao, Z., Bazmi, A. A., Abbas, T., & Yu, G. (2016). Progress in the production of biomass-to-liquid biofuels to decarbonize the transport sector – prospects and challenges. *RSC Advances*, 6(38), 32140–32170. <https://doi.org/10.1039/C5RA26459F>
- Binder, J. B., & Raines, R. T. (2009). Simple chemical transformation of lignocellulosic biomass into furans for fuels and chemicals. *Journal of the American Chemical Society*, 131(5), 1979–1985. <https://doi.org/10.1021/ja808537j>
- Bongaerts, J., Krämer, M., Müller, U., Raeven, L., & Wubbolts, M. (2001). Metabolic engineering for microbial production of aromatic amino acids and derived compounds. *Metabolic Engineering*, 3(4), 289–300. <https://doi.org/10.1006/mben.2001.0196>
- Bragin, G. E., Davis, C. W., Kung, M. H., Kelley, B. A., Sutherland, C. A., & Lampi, M. A. (2020). Biodegradation and ecotoxicity of branched alcohol ethoxylates: Application of the target lipid model and implications for environmental classification. *Journal of Surfactants and Detergents*, 23(2), 383–403. <https://doi.org/10.1002/jsde.12359>
- Brehmer, B. (2008). Chemical biorefinery perspectives : The valorisation of functionalised chemicals from biomass resources compared to the conventional fossil fuel production route.
- Brethauer, S., & Studer, M. H. (2014). Consolidated bioprocessing of lignocellulose by a microbial consortium. *Energy & Environmental Science*, 7(4), 1446–1453. <https://doi.org/10.1039/c3ee41753k>
- Brodeur, G., Yau, E., Badal, K., Collier, J., Ramachandran, K., & Ramakrishnan, S. (2011). Chemical and physicochemical pretreatment of lignocellulosic biomass: A review. *Enzyme Research*, 2011, 1–17. <https://doi.org/10.4061/2011/787532>
- Buchholz, J., Schwentner, A., Brunnenkan, B., Gabris, C., Grimm, S., Gerstmeir, R., Takors, R., Eikmanns, B. J., & Blombach, B. (2013). Platform engineering of *Corynebacterium glutamicum* with reduced pyruvate dehydrogenase complex activity for improved

- production of L-lysine, L-valine, and 2-ketoisovalerate. *Applied and Environmental Microbiology*, 79(18), 5566–5575. <https://doi.org/10.1128/AEM.01741-13>
- Bustamante, D., Segarra, S., Tortajada, M., Ramón, D., Del Cerro, C., Auxiliadora Prieto, M., Iglesias, J. R., & Rojas, A. (2019). *In silico* prospection of microorganisms to produce polyhydroxyalkanoate from whey: *Caulobacter segnis* DSM 29236 as a suitable industrial strain. *Microbial Biotechnology*, 12(3), 487–501. <https://doi.org/10.1111/1751-7915.13371>
- Calderón, C., Gauthier, G., & Jossart, J.-M. (2016). *AEBIOM Statistical report–European bioenergy outlook full report 2016*. AEBIOM.
- Capolupo, L., & Faraco, V. (2016). Green methods of lignocellulose pretreatment for biorefinery development. *Applied Microbiology and Biotechnology*, 100(22), 9451–9467. <https://doi.org/10.1007/s00253-016-7884-y>
- Cazetta, M. L., Celligoi, M. A. P. C., Buzato, J. B., Scarmino, I. S., & da Silva, R. S. F. (2005). Optimization study for sorbitol production by *Zymomonas mobilis* in sugar cane molasses. *Process Biochemistry*, 40(2), 747–751. <https://doi.org/10.1016/j.procbio.2004.01.041>
- Cesário, M. T., Raposo, R. S., de Almeida, M. C. M., van Keulen, F., Ferreira, B. S., & da Fonseca, M. M. R. (2014). Enhanced bioproduction of poly-3-hydroxybutyrate from wheat straw lignocellulosic hydrolysates. *New Biotechnology*, 31(1), 104–113. <https://doi.org/10.1016/j.nbt.2013.10.004>
- Chandrasekhar, K., Lee, Y.-J., & Lee, D.-W. (2015). Biohydrogen production: Strategies to improve process efficiency through microbial routes. *International Journal of Molecular Sciences*, 16(4), 8266–8293. <https://doi.org/10.3390/ijms16048266>
- Chen, C., Ding, S., Wang, D., Li, Z., & Ye, Q. (2014). Simultaneous saccharification and fermentation of cassava to succinic acid by *Escherichia coli* NZN111. *Bioresource Technology*, 163, 100–105. <https://doi.org/10.1016/j.biortech.2014.04.020>
- Cherubini, F., Jungmeier, G., Wellisch, M., Willke, T., Skiadas, I., Van Ree, R., & de Jong, E. (2009). Toward a common classification approach for biorefinery systems. *Biofuels, Bioproducts and Biorefining*, 3(5), 534–546. <https://doi.org/10.1002/bbb.172>
- Cho, K.-M., Choi, K.-o., & Bae, H.-a. (2013). Process for producing fermentation product from carbon sources containing glycerol using *Corynebacteria* USA.
- Choi, S., Song, C. W., Shin, J. H., & Lee, S. Y. (2015). Biorefineries for the production of top building block chemicals and their derivatives. *Metabolic Engineering*, 28, 223–239. <https://doi.org/10.1016/j.ymben.2014.12.007>
- Cimini, D., Argenzio, O., D'Ambrosio, S., Lama, L., Finore, I., Finamore, R., Pepe, O., Faraco, V., & Schiraldi, C. (2016). Production of succinic acid from *Basfia succiniciproducens* up to the pilot scale from *Arundo donax* hydrolysate. *Bioresource Technology*, 222, 355–360. <https://doi.org/10.1016/j.biortech.2016.10.004>
- Clark, J. H., Farmer, T. J., Hunt, A. J., & Sherwood, J. (2015). Opportunities for bio-based solvents created as petrochemical and fuel products transition towards renewable resources. *International Journal of Molecular Sciences*, 16(8), 17101–17159. <https://doi.org/10.3390/ijms160817101>
- Cleto, S., Jensen, J. V. K., Wendisch, V. F., & Lu, T. K. (2016). *Corynebacterium glutamicum* metabolic engineering with CRISPR interference (CRISPRi). *ACS Synthetic Biology*, 5(5), 375–385. <https://doi.org/10.1021/acssynbio.5b00216>
- Cok, B., Tsiropoulos, I., Roes, A. L., & Patel, M. K. (2014). Succinic acid production derived from carbohydrates: An energy and greenhouse gas assessment of a platform chemical toward a bio-based economy. *Biofuels, Bioproducts and Biorefining*, 8(1), 16–29. <https://doi.org/10.1002/bbb.1427>

- Coradini, A. L. V., Anschau, A., Vidotti, A. D. S., Reis, É. M., Xavier, MdCA., Coelho, R. S., & Franco, T. T. (2015). Microorganism for bioconversion of sugar hydrolysates into lipids. In *Microorganisms in Biorefineries* (pp. 51–78). Springer.
- Cordero-Ravelo, V., & Schallenberg-Rodriguez, J. (2018). Biodiesel production as a solution to waste cooking oil (WCO) disposal. Will any type of WCO do for a transesterification process? A quality assessment. *Journal of Environmental Management*, 228, 117–129. <https://doi.org/10.1016/j.jenvman.2018.08.106>
- Cuéllar, A. D., & Webber, M. E. (2008). Cow power: The energy and emissions benefits of converting manure to biogas. *Environmental Research Letters*, 3(3), 034002. <https://doi.org/10.1088/1748-9326/3/3/034002>
- Dahunsi, S., Olayanju, T., Izebere, J., & Oluyori, A. (2018). Data on energy and economic evaluation and microbial assessment of anaerobic co-digestion of fruit rind of *Telfairia occidentalis* (Fluted pumpkin) and poultry manure. *Data in Brief*, 21, 97–104. <https://doi.org/10.1016/j.dib.2018.09.065>
- Dasgupta, D., Bandhu, S., Adhikari, D. K., & Ghosh, D. (2017). Challenges and prospects of xylitol production with whole cell bio-catalysis: A review. *Microbiological Research*, 197, 9–21. <https://doi.org/10.1016/j.micres.2016.12.012>
- de Albuquerque, T. L., da Silva, I. J., Jr, de Macedo, G. R., & Rocha, M. V. P. (2014). Biotechnological production of xylitol from lignocellulosic wastes: A review. *Process Biochemistry*, 49(11), 1779–1789. <https://doi.org/10.1016/j.procbio.2014.07.010>
- de Jong, E., & Jungmeier, G. (2015). Biorefinery concepts in comparison to petrochemical refineries. In A. Pandey (Ed.), *Industrial Biorefineries and White Biotechnology* (pp. 3–33). Elsevier.
- de Kalbermatten, N., Ravussin, E., Maeder, E., Geser, C., Jéquier, E., & Felber, J. P. (1980). Comparison of glucose, fructose, sorbitol, and xylitol utilization in humans during insulin suppression. *Metabolism: Clinical and Experimental*, 29(1), 62–67. [https://doi.org/10.1016/0026-0495\(80\)90099-2](https://doi.org/10.1016/0026-0495(80)90099-2)
- Delrue, F., Setier, P. A., Sahut, C., Cournac, L., Roubaud, A., Peltier, G., & Froment, A. K. (2012). An economic, sustainability, and energetic model of biodiesel production from microalgae. *Bioresource Technology*, 111, 191–200. <https://doi.org/10.1016/j.biortech.2012.02.020>
- den Haan, R., van Rensburg, E., Rose, S. H., Görgens, J. F., & van Zyl, W. H. (2015). Progress and challenges in the engineering of non-cellulolytic microorganisms for consolidated bioprocessing. *Current Opinion in Biotechnology*, 33, 32–38. <https://doi.org/10.1016/j.copbio.2014.10.003>
- Dessbesell, L., Xu, C., Pulkki, R., Leitch, M., & Mahmood, N. (2017). Forest biomass supply chain optimization for a biorefinery aiming to produce high-value bio-based materials and chemicals from lignin and forestry residues: A review of literature. *Canadian Journal of Forest Research*, 47(3), 277–288. <https://doi.org/10.1139/cjfr-2016-0336>
- Devarapalli, M., & Atiyeh, H. K. (2015). A review of conversion processes for bioethanol production with a focus on syngas fermentation. *Biofuel Research Journal*, 2(3), 268–280. <https://doi.org/10.18331/BRJ2015.2.3.5>
- Díez-Pascual, A. M. (2019). *Synthesis and applications of biopolymer composites*. Multidisciplinary Digital Publishing Institute.
- Dimitriadis, A., Natsios, I., Dimaratos, A., Katsaounis, D., Samaras, Z., Bezergianni, S., & Lehto, K. (2018). Evaluation of a hydrotreated vegetable oil (HVO) and effects on emissions of a passenger car diesel engine. *Frontiers of Mechanical Engineering*, 4, 7.

- Du, G., Chen, L. X., & Yu, J. (2004). High-efficiency production of bioplastics from biodegradable organic solids. *Journal of Polymers and the Environment*, 12(2), 89–94. <https://doi.org/10.1023/B:JOOE.0000010054.58019.21>
- Dudley, B. (2018). BP statistical review of world energy. BP Statistical Review. accessed Aug 6, 2018.
- Dutta, S., De, S., Saha, B., & Alam, M. I. (2012). Advances in conversion of hemicellulosic biomass to furfural and upgrading to biofuels. *Catalysis Science & Technology*, 2(10), 2025–2036. <https://doi.org/10.1039/c2cy20235b>
- Eickhout, B. (2012). A strategy for a bio-based economy. *Green New Deal Series*, 9, 1–52.
- Elshout, P. M., van Zelm, R., van der Velde, M., Steinmann, Z., & Huijbregts, M. A. (2019). Global relative species loss due to first-generation biofuel production for the transport sector. *Global Change Biology. Bioenergy*, 11(6), 763–772. <https://doi.org/10.1111/gcbb.12597>
- Esteves, E. M. M., Herrera, A. M. N., Esteves, V. P. P., & Morgado, C. d. R. V. (2019). Life cycle assessment of manure biogas production: A review. *Journal of Cleaner Production*, 219, 411–423. <https://doi.org/10.1016/j.jclepro.2019.02.091>
- Eurostat (2019). *Development of the production of primary energy (by fuel type), EU-28, 2007-2017*. https://ec.europa.eu/eurostat/statistics-explained/index.php/Energy_production_and_imports#Production_of_primary_energy_decreased_between_2007_and_2017
- Fagbohungebe, M. O., Herbert, B. M. J., Hurst, L., Ibeto, C. N., Li, H., Usmani, S. Q., & Semple, K. T. (2017). The challenges of anaerobic digestion and the role of biochar in optimizing anaerobic digestion. *Waste Management (New York, N.Y.)*, 61, 236–249. <https://doi.org/10.1016/j.wasman.2016.11.028>
- FAO (2011). *The state of the world's land and water resources for food and agriculture*. FAO and Earthscan.
- FAO, F. (2014). Global forest products facts and figures. FAO Forestry Department Rome.
- Favaro, L., Jansen, T., & van Zyl, W. H. (2019). Exploring industrial and natural *Saccharomyces cerevisiae* strains for the bio-based economy from biomass: the case of bioethanol. *Critical Reviews in Biotechnology*, 39(6), 800–816. <https://doi.org/10.1080/07388551.2019.1619157>
- Fialho, A. M., Moreira, L. M., Granja, A. T., Popescu, A. O., Hoffmann, K., & Sá-Correia, I. (2008). Occurrence, production, and applications of gellan: Current state and perspectives. *Applied Microbiology and Biotechnology*, 79(6), 889–900. <https://doi.org/10.1007/s00253-008-1496-0>
- Fuso Nerini, F., Tomei, J., To, L. S., Bisaga, I., Parikh, P., Black, M., Borrión, A., Spataru, C., Castán Broto, V., Anandarajah, G., Milligan, B., & Mulugetta, Y. (2018). Mapping synergies and trade-offs between energy and the sustainable development goals. *Nature Energy*, 3(1), 10–15. <https://doi.org/10.1038/s41560-017-0036-5>
- Gaida, S. M., Liedtke, A., Jentges, A. H. W., Engels, B., & Jennewein, S. (2016). Metabolic engineering of *Clostridium cellulolyticum* for the production of n-butanol from crystalline cellulose. *Microbial Cell Factories*, 15, 6 <https://doi.org/10.1186/s12934-015-0406-2>
- Gardezi, S. A., Joseph, B., Prado, F., & Barbosa, A. (2013). Thermochemical biomass to liquid (BTL) process: Bench-scale experimental results and projected process economics of a commercial scale process. *Biomass and Bioenergy*, 59, 168–186. <https://doi.org/10.1016/j.biombioe.2013.09.010>
- George, S., & Jayachandran, K. (2013). Production and characterization of rhamnolipid bio-surfactant from waste frying coconut oil using a novel *Pseudomonas aeruginosa* D. *Journal of Applied Microbiology*, 114(2), 373–383. <https://doi.org/10.1111/jam.12069>

- Gewald, D., Siokos, K., Karellas, S., & Spliethoff, H. (2012). Waste heat recovery from a landfill gas-fired power plant. *Renewable and Sustainable Energy Reviews*, 16(4), 1779–1789. <https://doi.org/10.1016/j.rser.2012.01.036>
- Goldemberg, J. (2007). Ethanol for a sustainable energy future. *Science (New York, N.Y.)*, 315(5813), 808–810. <https://doi.org/10.1126/science.1137013>
- Gollakota, A. R. K., Kishore, N., & Gu, S. (2018). A review on hydrothermal liquefaction of biomass. *Renewable and Sustainable Energy Reviews*, 81, 1378–1392. <https://doi.org/10.1016/j.rser.2017.05.178>
- Gopinath, V., Meiswinkel, T. M., Wendisch, V. F., & Nampoothiri, K. M. (2011). Amino acid production from rice straw and wheat bran hydrolysates by recombinant pentose-utilizing *Corynebacterium glutamicum*. *Applied Microbiology and Biotechnology*, 92(5), 985–996. <https://doi.org/10.1007/s00253-011-3478-x>
- Gortari, M. C., & Hours, R. A. (2013). Biotechnological processes for chitin recovery out of crustacean waste: A mini-review. *Electronic Journal of Biotechnology*, 16, 14–14.
- Gowen, C. M., & Fong, S. S. (2011). Applications of systems biology towards microbial fuel production. *Trends in Microbiology*, 19(10), 516–524. <https://doi.org/10.1016/j.tim.2011.07.005>
- Green, E. M. (2011). Fermentative production of butanol-the industrial perspective. *Current Opinion in Biotechnology*, 22(3), 337–343. <https://doi.org/10.1016/j.copbio.2011.02.004>
- Grimberg, S. J., Hilderbrandt, D., Kinnunen, M., & Rogers, S. (2015). Anaerobic digestion of food waste through the operation of a mesophilic two-phase pilot scale digester-assessment of variable loadings on system performance. *Bioresource Technology*, 178, 226–229. <https://doi.org/10.1016/j.biortech.2014.09.001>
- Guirimand, G., Sasaki, K., Inokuma, K., Bamba, T., Hasunuma, T., & Kondo, A. (2016). Cell surface engineering of *Saccharomyces cerevisiae* combined with membrane separation technology for xylitol production from rice straw hydrolysate. *Applied Microbiology and Biotechnology*, 100(8), 3477–3487. <https://doi.org/10.1007/s00253-015-7179-8>
- Gustavsson, J., Cederberg, C., Sonesson, U., Van Otterdijk, R., & Meybeck, A. (2011). Global food losses and food waste. FAO Rome.
- Hamaguchi, M., Cardoso, M., & Vakkilainen, E. (2012). Alternative technologies for bio-fuels production in kraft pulp mills—potential and prospects. *Energies*, 5(7), 2288–2309. <https://doi.org/10.3390/en5072288>
- Hamme, V., Sannier, F., Piot, J.-M., Didelot, S., & Bordenave-Juchereau, S. (2009). Crude goat whey fermentation by *Kluyveromyces marxianus* and *Lactobacillus rhamnosus*: Contribution to proteolysis and ACE inhibitory activity. *The Journal of Dairy Research*, 76(2), 152–157. <https://doi.org/10.1017/S0022029908003877>
- Harith, Z. T., Charalampopoulos, D., & Chatzifragkou, A. (2019). Rapeseed meal hydrolysate as substrate for microbial astaxanthin production. *Biochemical Engineering Journal*, 151, 107330. <https://doi.org/10.1016/j.bej.2019.107330>
- Harnedy, P. A., & FitzGerald, R. J. (2012). Bioactive peptides from marine processing waste and shellfish: A review. *Journal of Functional Foods*, 4(1), 6–24. <https://doi.org/10.1016/j.jff.2011.09.001>
- He, J., Liu, M., Huang, K., Walker, T. W., Maravelias, C. T., Dumesic, J. A., & Huber, G. W. (2017). Production of levoglucosenone and 5-hydroxymethylfurfural from cellulose in polar aprotic solvent–water mixtures. *Green Chemistry*, 19(15), 3642–3653. <https://doi.org/10.1039/C7GC01688C>
- Higashide, W., Li, Y., Yang, Y., & Liao, J. C. (2011). Metabolic engineering of *Clostridium cellulolyticum* for production of isobutanol from cellulose. *Applied and Environmental Microbiology*, 77(8), 2727–2733. <https://doi.org/10.1128/AEM.02454-10>

- Himmel, M. E., Ding, S.-Y., Johnson, D. K., Adney, W. S., Nimlos, M. R., Brady, J. W., & Foust, T. D. (2007). Biomass recalcitrance: Engineering plants and enzymes for biofuels production. *Science (New York, N.Y.)*, 315(5813), 804–807. <https://doi.org/10.1126/science.1137016>
- Ho, S.-H., Chen, Y.-D., Chang, C.-Y., Lai, Y.-Y., Chen, C.-Y., Kondo, A., Ren, N.-Q., & Chang, J.-S. (2017). Feasibility of CO₂ mitigation and carbohydrate production by micro-alga *Scenedesmus obliquus* CNW-N used for bioethanol fermentation under outdoor conditions: Effects of seasonal changes. *Biotechnology for Biofuels*, 10, 27 <https://doi.org/10.1186/s13068-017-0712-5>
- Hood, E. E. (2016). Plant-based biofuels. *F1000Research*, 5, 185. <https://doi.org/10.12688/f1000research.7418.1>
- Hu, J., Lin, Y., Zhang, Z., Xiang, T., Mei, Y., Zhao, S., Liang, Y., & Peng, N. (2016). High-titer lactic acid production by *Lactobacillus pentosus* FL0421 from corn stover using fed-batch simultaneous saccharification and fermentation. *Bioresource Technology*, 214, 74–80. <https://doi.org/10.1016/j.biortech.2016.04.034>
- Hu, Q., Yang, H., Yao, D., Zhu, D., Wang, X., Shao, J., & Chen, H. (2016). The densification of bio-char: Effect of pyrolysis temperature on the qualities of pellets. *Bioresource Technology*, 200, 521–527. <https://doi.org/10.1016/j.biortech.2015.10.077>
- Hu, J., Zhang, Z., Lin, Y., Zhao, S., Mei, Y., Liang, Y., & Peng, N. (2015). High-titer lactic acid production from NaOH-pretreated corn stover by *Bacillus coagulans* LA204 using fed-batch simultaneous saccharification and fermentation under non-sterile condition. *Bioresource Technology*, 182, 251–257. <https://doi.org/10.1016/j.biortech.2015.02.008>
- Igartua, A., Fernández, X., Areitioaurtena, O., Luther, R., Seyfert, C., Rausch, J., Illarramendi, I., Berg, M., Schultheiß, H., Duffau, B., Plouseau, S., & Woydt, M. (2009). Biolubricants and triboreactive materials for automotive applications. *Tribology International*, 42(4), 561–568. <https://doi.org/10.1016/j.triboint.2008.10.015>
- Insam, H., Gómez-Brandón, M., & Ascher, J. (2015). Manure-based biogas fermentation residues – friend or foe of soil fertility? *Soil Biology and Biochemistry*, 84, 1–14. <https://doi.org/10.1016/j.soilbio.2015.02.006>
- Irmak, S. (2019). *Challenges of biomass utilization for biofuels Biomass for bioenergy-recent trends and future challenges*. IntechOpen.
- Isikgor, F. H., & Becer, C. R. (2015). Lignocellulosic biomass: A sustainable platform for the production of bio-based chemicals and polymers. *Polymer Chemistry*, 6(25), 4497–4559. <https://doi.org/10.1039/C5PY00263J>
- Jahirul, M. I., Rasul, M. G., Chowdhury, A. A., & Ashwath, N. (2012). Biofuels production through biomass pyrolysis —a technological review. *Energies*, 5(12), 4952–5001. <https://doi.org/10.3390/en5124952>
- Jansen, M. L. A., & van Gulik, W. M. (2014). Towards large scale fermentative production of succinic acid. *Current Opinion in Biotechnology*, 30, 190–197. <https://doi.org/10.1016/j.copbio.2014.07.003>
- Jeswani, H. K., Whiting, A., Martin, A., & Azapagic, A. (2019). Environmental and economic sustainability of poultry litter gasification for electricity and heat generation. *Waste Management (New York, N.Y.)*, 95, 182–191. <https://doi.org/10.1016/j.wasman.2019.05.053>
- Jezierska, S., Claus, S., Ledesma-Amaro, R., & Van Bogaert, I. (2019). Redirecting the lipid metabolism of the yeast *Starmerella bombicola* from glycolipid to fatty acid production. *Journal of Industrial Microbiology & Biotechnology*, 46(12), 1697–1706. <https://doi.org/10.1007/s10295-019-02234-x>

- Jin, H. S., Park, S. Y., Kim, J. Y., Lee, J. E., Lee, H. S., Kang, N. J., & Lee, D. W. (2019). Fluorescence-based quantification of bioactive keratin peptides from feathers for optimizing large-scale anaerobic fermentation and purification. *Biotechnology and Bioprocess Engineering*, 24(1), 240–249. <https://doi.org/10.1007/s12257-018-0400-8>
- Jin, H. S., Song, K., Baek, J. H., Lee, J. E., Kim, D. J., Nam, G. W., Kang, N. J., & Lee, D. W. (2018). Identification of matrix metalloproteinase-1-suppressive peptides in feather keratin hydrolysate. *Journal of Agricultural and Food Chemistry*, 66(48), 12719–12729. <https://doi.org/10.1021/acs.jafc.8b05213>
- John, R. P., Anisha, G., Nampoothiri, K. M., & Pandey, A. (2011). Micro and macroalgal biomass: A renewable source for bioethanol. *Bioresource Technology*, 102(1), 186–193. <https://doi.org/10.1016/j.biortech.2010.06.139>
- Jong, Ed., Higson, A., Walsh, P., & Wellisch, M. (2010). Bio-based chemicals value added products from biorefineries.
- Jönsson, L. J., & Martín, C. (2016). Pretreatment of lignocellulose: Formation of inhibitory by-products and strategies for minimizing their effects. *Bioresource Technology*, 199, 103–112. <https://doi.org/10.1016/j.biortech.2015.10.009>
- Jun, S.-A., Moon, C., Kang, C.-H., Kong, S. W., Sang, B.-I., & Um, Y. (2010). Microbial fed-batch production of 1,3-propanediol using raw glycerol with suspended and immobilized *Klebsiella pneumoniae*. *Applied Biochemistry and Biotechnology*, 161(1-8), 491–501. <https://doi.org/10.1007/s12010-009-8839-x>
- Jung, M.-Y., Ng, C. Y., Song, H., Lee, J., & Oh, M.-K. (2012). Deletion of lactate dehydrogenase in *Enterobacter aerogenes* to enhance 2,3-butanediol production. *Applied Microbiology and Biotechnology*, 95(2), 461–469. <https://doi.org/10.1007/s00253-012-3883-9>
- Kalogiannis, S., Iakovidou, G., Liakopoulou-Kyriakides, M., Kyriakidis, D. A., & Skaracis, G. N. (2003). Optimization of xanthan gum production by *Xanthomonas campestris* grown in molasses. *Process Biochemistry*, 39(2), 249–256. [https://doi.org/10.1016/S0032-9592\(03\)00067-0](https://doi.org/10.1016/S0032-9592(03)00067-0)
- Kang, N. J., Jin, H.-S., Lee, S.-E., Kim, H. J., Koh, H., & Lee, D.-W. (2020). New approaches towards the discovery and evaluation of bioactive peptides from natural resources. *Critical Reviews in Environmental Science and Technology*, 50(1), 72–103. <https://doi.org/10.1080/10643389.2019.1619376>
- Karatzos, S., van Dyk, J. S., McMillan, J. D., & Saddler, J. (2017). Drop-in biofuel production via conventional (lipid/fatty acid) and advanced (biomass) routes. Part I. *Biofuels, Bioproducts and Biorefining*, 11(2), 344–362. <https://doi.org/10.1002/bbb.1746>
- Karmee, S. K. (2016). Liquid biofuels from food waste: Current trends, prospect and limitation. *Renewable and Sustainable Energy Reviews*, 53, 945–953. <https://doi.org/10.1016/j.rser.2015.09.041>
- Kayhanian, M. (1995). Biodegradability of the organic fraction of municipal solid-waste in a high-solids anaerobic digester. *Waste Management & Research*, 13(2), 123–136. <https://doi.org/10.1177/0734242X9501300203>
- Keasling, J. D. (2010). Manufacturing molecules through metabolic engineering. *Science (New York, N.Y.)*, 330(6009), 1355–1358. <https://doi.org/10.1126/science.1193990>
- Kenny, S. T., Runic, J. N., Kaminsky, W., Woods, T., Babu, R. P., & O'Connor, K. E. (2012). Development of a bioprocess to convert PET derived terephthalic acid and bio-diesel derived glycerol to medium chain length polyhydroxyalkanoate. *Applied Microbiology and Biotechnology*, 95(3), 623–633. <https://doi.org/10.1007/s00253-012-4058-4>

- Khanafari, A., Marandi, R., & Sanati, S. (2008). Recovery of chitin and chitosan from shrimp waste by chemical and microbial methods. *Journal of Environmental Health Science and Engineering*.
- Kim, S-h., & Han, G.-S. (2016). Fuel characteristics of biomass pellets fabricated with reed stalk. *Journal of Korea Technical Association of the Pulp and Paper Industry*, 48(3), 99–106. <https://doi.org/10.7584/ktappi.2016.48.3.099>
- Kimura, Y. (2009). Molecular, structural, and material design of bio-based polymers. *Polymer Journal*, 41(10), 797–807. <https://doi.org/10.1295/polymj.PJ2009154>
- Kirchner, O., & Tauch, A. (2003). Tools for genetic engineering in the amino acid-producing bacterium *Corynebacterium glutamicum*. *Journal of Biotechnology*, 104(1-3), 287–299. [https://doi.org/10.1016/S0168-1656\(03\)00148-2](https://doi.org/10.1016/S0168-1656(03)00148-2)
- Kitamoto, D., Isoda, H., & Nakahara, T. (2002). Functions and potential applications of glycolipid biosurfactants — from energy-saving materials to gene delivery carriers. *Journal of Bioscience and Bioengineering*, 94(3), 187–201. [https://doi.org/10.1016/S1389-1723\(02\)80149-9](https://doi.org/10.1016/S1389-1723(02)80149-9)
- Koller, M. (2015). Recycling of waste streams of the biotechnological poly (hydroxyalkanoate) production by *Haloferax mediterranei* on whey. *International Journal of Polymer Science*, 2015, 1–8. <https://doi.org/10.1155/2015/370164>
- Komesu, A., Oliveira, J., da Silva Martins, L. H., Maciel, M. R. W., & Filho, R. M. (2018). *Lactic acid and ethanol: Promising bio-based chemicals from fermentation*, 87–116). Wiley Online Library.
- Kotarska, K., Dziemianowicz, W., & Świerczyńska, A. (2019). Study on the sequential combination of bioethanol and biogas production from corn straw. *Molecules*, 24(24), 4558. <https://doi.org/10.3390/molecules24244558>
- Koti, S., Govumoni, S. P., Gentela, J., & Venkateswar Rao, L. (2016). Enhanced bioethanol production from wheat straw hemicellulose by mutant strains of pentose fermenting organisms *Pichia stipitis* and *Candida shehatae*. *SpringerPlus*, 5(1), 1545–1545. <https://doi.org/10.1186/s40064-016-3222-1>
- Koutinas, A. A., Vlysidis, A., Pleissner, D., Kopsahelis, N., Lopez Garcia, I., Kookos, I. K., Papanikolaou, S., Kwan, T. H., & Lin, C. S. K. (2014). Valorization of industrial waste and by-product streams via fermentation for the production of chemicals and biopolymers. *Chemical Society Reviews*, 43(8), 2587–2627. <https://doi.org/10.1039/c3cs60293a>
- Kretschmer, U., Kirchner, N., Morhart, C., & Spiecker, H. (2013). A new approach to assessing tree stem quality characteristics using terrestrial laser scans. *Silva Fennica*, 47(5) <https://doi.org/10.14214/sf.1071>
- Kumar, A., Demirel, Y., Jones, D. D., & Hanna, M. A. (2010). Optimization and economic evaluation of industrial gas production and combined heat and power generation from gasification of corn stover and distillers grains. *Bioresource Technology*, 101(10), 3696–3701. <https://doi.org/10.1016/j.biortech.2009.12.103>
- Kuo, Y.-C., Yuan, S.-F., Wang, C.-A., Huang, Y.-J., Guo, G.-L., & Hwang, W.-S. (2015). Production of optically pure L-lactic acid from lignocellulosic hydrolysate by using a newly isolated and D-lactate dehydrogenase gene-deficient *Lactobacillus paracasei* strain. *Bioresource Technology*, 198, 651–657. <https://doi.org/10.1016/j.biortech.2015.09.071>
- Lantz, M. (2012). The economic performance of combined heat and power from biogas produced from manure in Sweden—a comparison of different CHP technologies. *Applied Energy*, 98, 502–511. <https://doi.org/10.1016/j.apenergy.2012.04.015>
- Lee, S. K., Chou, H., Ham, T. S., Lee, T. S., & Keasling, J. D. (2008). Metabolic engineering of microorganisms for biofuels production: From bugs to synthetic biology to fuels. *Curr Opin Biotechnol*, 19(6), 556–563. <https://doi.org/10.1016/j.copbio.2008.10.014>

- Lee, S. Y., & Kim, H. U. (2015). Systems strategies for developing industrial microbial strains. *Nature Biotechnology*, 33(10), 1061–1072. <https://doi.org/10.1038/nbt.3365>
- Lee, D. W., & Lee, S. J. (2015). Microbial platform cells for synthetic biology. In A. Glieder (Ed.), *Synthetic biology* (pp. 229–254). Springer.
- Lee, S. J., Lee, S. J., & Lee, D. W. (2013). Design and development of synthetic microbial platform cells for bioenergy. *Frontiers in Microbiology*, 4, 92 <https://doi.org/10.3389/fmicb.2013.00092>
- Lemes, A. C., Sala, L., Ores, JdC., Braga, A. R. C., Egea, M. B., & Fernandes, K. F. (2016). A review of the latest advances in encrypted bioactive peptides from protein-rich waste. *International Journal of Molecular Sciences*, 17(6), 950. <https://doi.org/10.3390/ijms17060950>
- Leung, C. C. J., Cheung, A. S. Y., Zhang, A. Y.-Z., Lam, K. F., & Lin, C. S. K. (2012). Utilisation of waste bread for fermentative succinic acid production. *Biochemical Engineering Journal*, 65, 10–15. <https://doi.org/10.1016/j.bej.2012.03.010>
- Lewandowski, I. (2015). Securing a sustainable biomass supply in a growing bioeconomy. *Global Food Security*, 6, 34–42. <https://doi.org/10.1016/j.gfs.2015.10.001>
- Liao, J. C., Mi, L., Pontrelli, S., & Luo, S. (2016). Fuelling the future: Microbial engineering for the production of sustainable biofuels. *Nature Reviews Microbiology*, 14(5), 288–304. <https://doi.org/10.1038/nrmicro.2016.32>
- Li, M., Meng, X., Diao, E., & Du, F. (2012). Xylitol production by *Candida tropicalis* from corn cob hemicellulose hydrolysate in a two-stage fed-batch fermentation process. *Journal of Chemical Technology & Biotechnology*, 87(3), 387–392. <https://doi.org/10.1002/jctb.2732>
- Li, S.-Y., Ng, I. S., Chen, P. T., Chiang, C.-J., & Chao, Y.-P. (2018). Biorefining of protein waste for production of sustainable fuels and chemicals. *Biotechnology for Biofuels*, 11, 256 <https://doi.org/10.1186/s13068-018-1234-5>
- Lin, C. S. K., Pfaltzgraff, L. A., Herrero-Davila, L., Mubofu, E. B., Abderrahim, S., Clark, J. H., Koutinas, A. A., Kopsahelis, N., Stamatelatou, K., Dickson, F., Thankappan, S., Mohamed, Z., Brocklesby, R., & Luque, R. (2013). Food waste as a valuable resource for the production of chemicals, materials and fuels. Current situation and global perspective. *Energy & Environmental Science*, 6(2), 426–464. <https://doi.org/10.1039/c2ee23440h>
- Liu, B., & Rajagopal, D. (2019). Life-cycle energy and climate benefits of energy recovery from wastes and biomass residues in the United States. *Nature Energy*, 4(8), 700–708. <https://doi.org/10.1038/s41560-019-0430-2>
- Li, K., Wang, B., Bolatibieke, D., Nan, D.-H., Zhang, Z.-X., Cui, M.-S., & Lu, Q. (2020). Catalytic fast pyrolysis of biomass with Ni-P-MCM-41 to selectively produce levoglucosone. *Journal of Analytical and Applied Pyrolysis*, 148, 104824. <https://doi.org/10.1016/j.jaap.2020.104824>
- Long, J.-Y., Song, K.-L., He, X., Zhang, B., Cui, X.-F., & Song, C.-F. (2018). Mutagenesis of PhaR, a regulator gene of polyhydroxyalkanoate biosynthesis of *Xanthomonas oryzae* pv. *oryzae* caused pleiotropic phenotype changes. *Frontiers in Microbiology*, 9 <https://doi.org/10.3389/fmicb.2018.03046>
- Luque, R., Herrero-Davila, L., Campelo, J. M., Clark, J. H., Hidalgo, J. M., Luna, D., Marinas, J. M., & Romero, A. A. (2008). Biofuels: a technological perspective. *Energy & Environmental Science*, 1(5), 542–564. <https://doi.org/10.1039/b807094f>
- Ma, K., Hu, G., Pan, L., Wang, Z., Zhou, Y., Wang, Y., Ruan, Z., & He, M. (2016). Highly efficient production of optically pure l-lactic acid from corn stover hydrolysate by thermophilic *Bacillus coagulans*. *Bioresource Technology*, 219, 114–122. <https://doi.org/10.1016/j.biortech.2016.07.100>

- Maity, S. K. (2015). Opportunities, recent trends and challenges of integrated biorefinery: Part I. *Renewable and Sustainable Energy Reviews*, 43, 1427–1445. <https://doi.org/10.1016/j.rser.2014.11.092>
- Majidian, P., Tabatabaei, M., Zeinolabedini, M., Naghshbandi, M. P., & Chisti, Y. (2018). Metabolic engineering of microorganisms for biofuel production. *Renewable and Sustainable Energy Reviews*, 82, 3863–3885. <https://doi.org/10.1016/j.rser.2017.10.085>
- Makkar, R. S., & Cameotra, S. S. (1999). Biosurfactant production by microorganisms on unconventional carbon sources. *Journal of Surfactants and Detergents*, 2(2), 237–241. <https://doi.org/10.1007/s11743-999-0078-3>
- Makkar, R. S., Cameotra, S. S., & Banat, I. M. (2011). Advances in utilization of renewable substrates for biosurfactant production. *AMB Express*, 1(1), 5 <https://doi.org/10.1186/2191-0855-1-5>
- Malico, I., Nepomuceno Pereira, R., Gonçalves, A. C., & Sousa, A. M. O. (2019). Current status and future perspectives for energy production from solid biomass in the European industry. *Renewable and Sustainable Energy Reviews*, 112, 960–977. <https://doi.org/10.1016/j.rser.2019.06.022>
- Mallepally, R. R., Marin, M. A., Montesdeoca, N., Parrish, C., Ward, K. R., & McHugh, M. A. (2015). Hydrogen peroxide loaded cellulose acetate mats as controlled topical O₂ delivery devices. *The Journal of Supercritical Fluids*, 105, 77–83. <https://doi.org/10.1016/j.supflu.2014.12.029>
- Maragkaki, A., Fountoulakis, M., Kyriakou, A., Lasaridi, K., & Manios, T. (2018). Boosting biogas production from sewage sludge by adding small amount of agro-industrial by-products and food waste residues. *Waste Manag*, 71, 605–611. <https://doi.org/10.1016/j.wasman.2017.04.024>
- Marchant, R., & Banat, I. M. (2012). Biosurfactants: A sustainable replacement for chemical surfactants? *Biotechnology Letters*, 34(9), 1597–1605. <https://doi.org/10.1007/s10529-012-0956-x>
- Marinho, G. S., Alvarado-Morales, M., & Angelidaki, I. (2016). Valorization of macroalga *Saccharina latissima* as novel feedstock for fermentation-based succinic acid production in a biorefinery approach and economic aspects. *Algal Research*, 16, 102–109. <https://doi.org/10.1016/j.algal.2016.02.023>
- Market Research Future. (2019). *Bio-based chemicals market: Information by type (bioplastics, bio-lubricants, bio-solvents, bio-alcohols, bio-based acids, bio-surfactants), application (food & beverage, pharmaceuticals, agriculture, packaging), and region — global forecast till 2023*. <https://www.marketresearchfuture.com/reports/bio-based-chemicals-market-5706>
- Marques, C., Tarek, R., Sara, M., & Brar, S. K. (2016). Chapter 12 - Sorbitol production from biomass and its global market. In S. Kaur Brar (Ed.), *Platform Chemical Biorefinery* (pp. 217–227). Elsevier.
- Masan, V., Burg, P., Visacki, V., & Ondrej, P. (2018). Evaluation of fuel pellets as secondary product when pressing oil from grapevine seeds. In *Proceedings 17th International Scientific Conference "Engineering for rural development, 23-25 May 2018* (pp. 1766–1770). Latvia University of Life Sciences and Technologies.
- Mata-Alvarez, J., Dosta, J., Romero-Güiza, M. S., Fonoll, X., Peces, M., & Astals, S. (2014). A critical review on anaerobic co-digestion achievements between 2010 and 2013. *Renewable and Sustainable Energy Reviews*, 36, 412–427. <https://doi.org/10.1016/j.rser.2014.04.039>

- Maurya, D. P., Singla, A., & Negi, S. (2015). An overview of key pretreatment processes for biological conversion of lignocellulosic biomass to bioethanol. 3 *Biotech*, 5(5), 597–609. <https://doi.org/10.1007/s13205-015-0279-4>
- Mazurkiewicz, J., Marczuk, A., Pochwatka, P., & Kujawa, S. (2019). Maize straw as a valuable energetic material for biogas plant feeding. *Materials*, 12(23), 3848. <https://doi.org/10.3390/ma12233848>
- McKendry, P. (2002). Energy production from biomass (part 1): Overview of biomass. *Bioresource Technology*, 83(1), 37–46. [https://doi.org/10.1016/S0960-8524\(01\)00118-3](https://doi.org/10.1016/S0960-8524(01)00118-3)
- Meiswinkel, T. M., Gopinath, V., Lindner, S. N., Nampoothiri, K. M., & Wendisch, V. F. (2013). Accelerated pentose utilization by *Corynebacterium glutamicum* for accelerated production of lysine, glutamate, ornithine and putrescine. *Microbial Biotechnology*, 6(2), 131–140. <https://doi.org/10.1111/1751-7915.12001>
- Melero, J. A., Iglesias, J., & Garcia, A. (2012). Biomass as renewable feedstock in standard refinery units. Feasibility, opportunities and challenges. *Energy & Environmental Science*, 5(6), 7393–7420. <https://doi.org/10.1039/c2ee21231e>
- Menegazzo, F., Ghedini, E., & Signoretto, M. (2018). 5-Hydroxymethylfurfural (HMF) production from real biomasses. *Molecules*, 23(9), 2201. <https://doi.org/10.3390/molecules23092201>
- Mesomo, M., Silva, M. F., Boni, G., Padilha, F. F., Mazutti, M., Mossi, A., de Oliveira, D., Cansian, R. L., Di Luccio, M., & Treichel, H. (2009). Xanthan gum produced by *Xanthomonas campestris* from cheese whey: Production optimisation and rheological characterisation. *Journal of the Science of Food and Agriculture*, 89(14), 2440–2445. <https://doi.org/10.1002/jsfa.3743>
- Metsoviti, M., Paramithiotis, S., Drosinos, E. H., Galiotou-Panayotou, M., Nychas, G. J. E., Zeng, A. P., & Papanikolaou, S. (2012). Screening of bacterial strains capable of converting biodiesel-derived raw glycerol into 1, 3-propanediol, 2, 3-butanediol and ethanol. *Engineering in Life Sciences*, 12(1), 57–68. <https://doi.org/10.1002/elsc.201100058>
- Metsoviti, M., Zeng, A.-P., Koutinas, A. A., & Papanikolaou, S. (2013). Enhanced 1,3-propanediol production by a newly isolated *Citrobacter freundii* strain cultivated on biodiesel-derived waste glycerol through sterile and non-sterile bioprocesses. *Journal of Biotechnology*, 163(4), 408–418. <https://doi.org/10.1016/j.jbiotec.2012.11.018>
- Mielenz, J. R. (2011). Biofuels from protein. *Nature Biotechnology*, 29(4), 327–328. <https://doi.org/10.1038/nbt.1838>
- Milovanovic, S., Markovic, D., Aksentijevic, K., Stojanovic, D. B., Ivanovic, J., & Zizovic, I. (2016). Application of cellulose acetate for controlled release of thymol. *Carbohydrate Polymers*, 147, 344–353. <https://doi.org/10.1016/j.carbpol.2016.03.093>
- Milovanovic, S., Stamenic, M., Markovic, D., Ivanovic, J., & Zizovic, I. (2015). Supercritical impregnation of cellulose acetate with thymol. *The Journal of Supercritical Fluids*, 97, 107–115. <https://doi.org/10.1016/j.supflu.2014.11.011>
- Mimitsuka, T., Sawai, H., Hatsu, M., & Yamada, K. (2007). Metabolic engineering of *Corynebacterium glutamicum* for cadaverine fermentation. *Bioscience, Biotechnology, and Biochemistry*, 71(9), 2130–2135. <https://doi.org/10.1271/bbb.60699>
- Mohr, A., & Raman, S. (2013). Lessons from first generation biofuels and implications for the sustainability appraisal of second generation biofuels. *Energy Policy*, 63(100), 114–122. <https://doi.org/10.1016/j.enpol.2013.08.033>
- Moshi, A. P., Temu, S. G., Nges, I. A., Malmo, G., Hosea, K. M. M., Elisante, E., & Mattiasson, B. (2015). Combined production of bioethanol and biogas from peels of wild cassava *Manihot glaziovii*. *Chemical Engineering Journal*, 279, 297–306. <https://doi.org/10.1016/j.cej.2015.05.006>

- Możejko-Ciesielska, J., & Serafim, L. S. (2019). Proteomic response of *Pseudomonas putida* KT2440 to dual carbon-phosphorus limitation during mcl-PHAs synthesis. *Biomolecules*, 9(12), 796. <https://doi.org/10.3390/biom9120796>
- Munasinghe, P. C., & Khanal, S. K. (2010). Biomass-derived syngas fermentation into bio-fuels: Opportunities and challenges. *Bioresource Technology*, 101(13), 5013–5022. <https://doi.org/10.1016/j.biortech.2009.12.098>
- Murakami, N., Oba, M., Iwamoto, M., Tashiro, Y., Noguchi, T., Bonkohara, K., Abdel-Rahman, M. A., Zendo, T., Shimoda, M., Sakai, K., & Sonomoto, K. (2016). L-Lactic acid production from glycerol coupled with acetic acid metabolism by *Enterococcus faecalis* without carbon loss. *Journal of Bioscience and Bioengineering*, 121(1), 89–95. <https://doi.org/10.1016/j.jbiosc.2015.05.009>
- Mussatto, S. I., & Dragone, G. (2016). Biomass pretreatment, biorefineries, and potential products for a bioeconomy development. In *Biomass fractionation technologies for a lignocellulosic feedstock based biorefinery* (pp. 1–22). Elsevier.
- Mussatto, S. I., & Teixeira, J. A. (2010). Increase in the fructooligosaccharides yield and productivity by solid-state fermentation with *Aspergillus japonicus* using agro-industrial residues as support and nutrient source. *Biochemical Engineering Journal*, 53(1), 154–157. <https://doi.org/10.1016/j.bej.2010.09.012>
- Nakajima, H., Dijkstra, P., & Loos, K. (2017). The recent developments in biobased polymers toward general and engineering applications: Polymers that are upgraded from biodegradable polymers, analogous to petroleum-derived polymers, and newly developed. *Polymers*, 9(12), 523. <https://doi.org/10.3390/polym9100523>
- Nakamura, C. E., & Whited, G. M. (2003). Metabolic engineering for the microbial production of 1,3-propanediol. *Current Opinion in Biotechnology*, 14(5), 454–459. <https://doi.org/10.1016/j.copbio.2003.08.005>
- Neuner, A., Wagner, I., Sieker, T., Ulber, R., Schneider, K., Peifer, S., & Heinzle, E. (2013). Production of l-lysine on different silage juices using genetically engineered *Corynebacterium glutamicum*. *Journal of Biotechnology*, 163(2), 217–224. <https://doi.org/10.1016/j.jbiotec.2012.07.190>
- Nitschke, M., Costa, S. G., Haddad, R., Gonçalves, L. A., Eberlin, M. N., & Contiero, J. (2005). Oil wastes as unconventional substrates for rhamnolipid biosurfactant production by *Pseudomonas aeruginosa* LBI. *Biotechnology Progress*, 21(5), 1562–1566. <https://doi.org/10.1021/bp050198x>
- nova-Institute (2019). Bioplastics market data. <https://www.european-bioplastics.org/market/>
- Nurfarahin, A. H., Mohamed, M. S., & Phang, L. Y. (2018). Culture medium development for microbial-derived surfactants production-an overview. *Molecules*, 23(5), 1049. <https://doi.org/10.3390/molecules23051049>
- Ögmundarson, Ó., Herrgård, M. J., Forster, J., Hauschild, M. Z., & Fantke, P. (2020). Addressing environmental sustainability of biochemicals. *Nature Sustainability*, 3(3), 167–174. <https://doi.org/10.1038/s41893-019-0442-8>
- Olajuyin, A. M., Yang, M., Liu, Y., Mu, T., Tian, J., Adaramoye, O. A., & Xing, J. (2016). Efficient production of succinic acid from *Palmaria palmata* hydrolysate by metabolically engineered *Escherichia coli*. *Bioresource Technology*, 214, 653–659. <https://doi.org/10.1016/j.biortech.2016.04.117>
- Olson, D. G., McBride, J. E., Joe Shaw, A., & Lynd, L. R. (2012). Recent progress in consolidated bioprocessing. *Current Opinion in Biotechnology*, 23(3), 396–405. <https://doi.org/10.1016/j.copbio.2011.11.026>

- Ortiz, M. E., Bleckwedel, J., Raya, R. R., & Mozzi, F. (2013). Biotechnological and *in situ* food production of polyols by lactic acid bacteria. *Applied Microbiology and Biotechnology*, 97(11), 4713–4726. <https://doi.org/10.1007/s00253-013-4884-z>
- Owusu, P. A., & Asumadu-Sarkodie, S. (2016). A review of renewable energy sources, sustainability issues and climate change mitigation. *Cogent Engineering*, 3(1), 1167990. <https://doi.org/10.1080/23311916.2016.1167990>
- Ozdal, M., & Başaran Kurbanoglu, E. (2019). Use of chicken feather peptone and sugar beet molasses as low cost substrates for xanthan production by *Xanthomonas campestris* MO-03. *Fermentation*, 5(1), 9. <https://doi.org/10.3390/fermentation5010009>
- Pacala, S., & Socolow, R. (2004). Stabilization wedges: Solving the climate problem for the next 50 years with current technologies. *Science (New York, N.Y.)*, 305(5686), 968–972. <https://doi.org/10.1126/science.1100103>
- Palaniraj, A., & Jayaraman, V. (2011). Production, recovery and applications of xanthan gum by *Xanthomonas campestris*. *Journal of Food Engineering*, 106(1), 1–12. <https://doi.org/10.1016/j.jfoodeng.2011.03.035>
- Panesar, P., Bali, V., Kumari, S., Babbar, N., & Oberoi, H. S. (2013). Prebiotics. In S. K. Brar. (Ed.) *Biotransformation of waste biomass into high value biochemicals* (pp. 237–259). Springer Science & Business Media.
- Pap, N., Pongrácz, E., Myllykoski, L., Keiski, R. (2004). *Waste Minimization and Utilization in the Food Industry: Processing of Arctic Berries, and Extraction of Valuable Compounds from Juice-Processing by-Products Proceedings of the Waste Minimization and Resources Use Optimization Conference, 10 June 2004* (pp. 159–168). Oulu University Press Oulu.
- Papadopoulou, E. L., Basnett, P., Paul, U. C., Marras, S., Ceseracciu, L., Roy, I., & Athanassiou, A. (2019). Green composites of poly (3-hydroxybutyrate) containing graphene nanoplatelets with desirable electrical conductivity and oxygen barrier properties. *ACS Omega*, 4(22), 19746–19755. <https://doi.org/10.1021/acsomega.9b02528>
- Paritosh, K., Yadav, M., Mathur, S., Balan, V., Liao, W., Pareek, N., & Vivekanand, V. (2018). Organic fraction of municipal solid waste: Overview of treatment methodologies to enhance anaerobic biodegradability. *Frontiers in Energy Research*, 6 <https://doi.org/10.3389/fenrg.2018.00075>
- Passoth, V., & Sandgren, M. (2019). Biofuel production from straw hydrolysates: Current achievements and perspectives. *Applied Microbiology and Biotechnology*, 103(13), 5105–5116. <https://doi.org/10.1007/s00253-019-09863-3>
- Peng, L., Xie, N., Guo, L., Wang, L., Yu, B., & Ma, Y. (2014). Efficient open fermentative production of polymer-grade L-lactate from sugarcane bagasse hydrolysate by thermotolerant *Bacillus* sp. strain P38. *PLoS One*, 9(9), e107143 <https://doi.org/10.1371/journal.pone.0107143>
- Pereira, S. C., Maehara, L., Machado, C. M. M., & Farinas, C. S. (2015). 2G ethanol from the whole sugarcane lignocellulosic biomass. *Biotechnology for Biofuels*, 8, 44 <https://doi.org/10.1186/s13068-015-0224-0>
- Picardo, A., Soltero, V. M., Peralta, M. E., & Chacartegui, R. (2019). District heating based on biogas from wastewater treatment plant. *Energy*, 180, 649–664. <https://doi.org/10.1016/j.energy.2019.05.123>
- Piotrowski, S., Carus, M., & Essel, R. (2016). Sustainable biomass supply and demand: A scenario analysis. *Open Agriculture*, 1(1) <https://doi.org/10.1515/opag-2016-0003>
- Qian, Z. G., Xia, X. X., & Lee, S. Y. (2009). Metabolic engineering of *Escherichia coli* for the production of putrescine: A four carbon diamine. *Biotechnology and Bioengineering*, 104(4), 651–662. <https://doi.org/10.1002/bit.22502>

- Quinchia, L., Delgado, M. A., Valencia, C., Franco, J., & Gallegos, C. (2009). Viscosity modification of high-oleic sunflower oil with polymeric additives for the design of new biolubricant formulations. *Environmental Science & Technology*, 43(6), 2060–2065. <https://doi.org/10.1021/es803047m>
- Radhika, D., & Murugesan, A. (2012). Bioproduction, statistical optimization and characterization of microbial plastic (poly 3-hydroxy butyrate) employing various hydrolysates of water hyacinth (*Eichhornia crassipes*) as sole carbon source. *Bioresource Technology*, 121, 83–92. <https://doi.org/10.1016/j.biortech.2012.06.107>
- Ragauskas, A. J., Williams, C. K., Davison, B. H., Britovsek, G., Cairney, J., Eckert, C. A., Frederick, W. J., Hallett, J. P., Leak, D. J., Liotta, C. L., Mielenz, J. R., Murphy, R., Templer, R., & Tschaplinski, T. (2006). The path forward for biofuels and biomaterials. *Science (New York, N.Y.)*, 311(5760), 484–489. <https://doi.org/10.1126/science.1114736>
- Reza, M. T., Lynam, J. G., Vasquez, V. R., & Coronella, C. J. (2012). Pelletization of biochar from hydrothermally carbonized wood. *Environmental Progress & Sustainable Energy*, 31(2), 225–234. <https://doi.org/10.1002/ep.11615>
- Rigouin, C., Lajus, S., Ocando, C., Borsenberger, V., Nicaud, J. M., Marty, A., Avérous, L., & Bordes, F. (2019). Production and characterization of two medium-chain-length poly-droxyalkanoates by engineered strains of *Yarrowia lipolytica*. *Microbial Cell Factories*, 18(1), 99 <https://doi.org/10.1186/s12934-019-1140-y>
- Rittmann, D., Lindner, S. N., & Wendisch, V. F. (2008). Engineering of a glycerol utilization pathway for amino acid production by *Corynebacterium glutamicum*. *Applied and Environmental Microbiology*, 74(20), 6216–6222. <https://doi.org/10.1128/AEM.00963-08>
- Ro, K. S., Cantrell, K. B., & Hunt, P. G. (2010). High-temperature pyrolysis of blended animal manures for producing renewable energy and value-added biochar. *Industrial & Engineering Chemistry Research*, 49(20), 10125–10131. <https://doi.org/10.1021/ie101155m>
- Robak, K., & Balcerek, M. (2018). Review of second generation bioethanol production from residual biomass. *Food Technology and Biotechnology*, 56(2), 174–187. <https://doi.org/10.17113/ftb.56.02.18.5428>
- Roesijadi, G., Jones, S. B., Snowden-Swan, L. J., & Zhu, Y. (2010). Macroalgae as a biomass feedstock: A preliminary analysis.
- Romero-Güiza, M. S., Vila, J., Mata-Alvarez, J., Chimenos, J. M., & Astals, S. (2016). The role of additives on anaerobic digestion: A review. *Renewable and Sustainable Energy Reviews*, 58, 1486–1499. <https://doi.org/10.1016/j.rser.2015.12.094>
- Rosales-Calderon, O., & Arantes, V. (2019). A review on commercial-scale high-value products that can be produced alongside cellulosic ethanol. *Biotechnology for Biofuels*, 12, 240 <https://doi.org/10.1186/s13068-019-1529-1>
- Ruiz, H. A., Rodríguez-Jasso, R. M., Fernandes, B. D., Vicente, A. A., & Teixeira, J. A. (2013). Hydrothermal processing, as an alternative for upgrading agriculture residues and marine biomass according to the biorefinery concept: A review. *Renewable and Sustainable Energy Reviews*, 21, 35–51. <https://doi.org/10.1016/j.rser.2012.11.069>
- Rujnić-Sokele, M., & Pilipović, A. (2017). Challenges and opportunities of biodegradable plastics: A mini review. *Waste Management & Research: The Journal of the International Solid Wastes and Public Cleansing Association, ISWA*, 35(2), 132–140. <https://doi.org/10.1177/0734242X16683272>
- Saini, J. K., Saini, R., & Tewari, L. (2015). Lignocellulosic agriculture wastes as biomass feedstocks for second-generation bioethanol production: Concepts and recent developments. *3 Biotech*, 5(4), 337–353. <https://doi.org/10.1007/s13205-014-0246-5>
- Salassi, M. E., Holzapfel, A., Hilbun, B. M., Deliberto, M. A., Gravois, K. A., Viator, H. P., Falconer, L. L., & Mark, T. B. (2017). Feedstock crop production costs and biofuel

- feedstock input costs associated with the production of energy cane and sweet sorghum in the Southeastern USA. *BioEnergy Research*, 10(3), 772–782. <https://doi.org/10.1007/s12155-017-9838-3>
- Salih, N., Salimon, J., Yousif, E., & Abdullah, B. M. (2013). Biolubricant basestocks from chemically modified plant oils: Ricinoleic acid based-tetraesters. *Chemistry Central Journal*, 7(1), 128 <https://doi.org/10.1186/1752-153X-7-128>
- Salvachúa, D., Mohagheghi, A., Smith, H., Bradfield, M. F. A., Nicol, W., Black, B. A., Biddy, M. J., Dowe, N., & Beckham, G. T. (2016). Succinic acid production on xylose-enriched biorefinery streams by *Actinobacillus succinogenes* in batch fermentation. *Biotechnology for Biofuels*, 9, 28 <https://doi.org/10.1186/s13068-016-0425-1>
- Salvachúa, D., Smith, H., St. John, P. C., Mohagheghi, A., Peterson, D. J., Black, B. A., Dowe, N., & Beckham, G. T. (2016). Succinic acid production from lignocellulosic hydrolysate by *Basfia succiniciproducens*. *Bioresource Technology*, 214, 558–566. <https://doi.org/10.1016/j.biortech.2016.05.018>
- Salvador, M., Abdulmutalib, U., Gonzalez, J., Kim, J., Smith, A. A., Faulon, J.-L., Wei, R., Zimmermann, W., & Jimenez, J. I. (2019). Microbial genes for a circular and sustainable bio-PET economy. *Genes*, 10(5), 373. <https://doi.org/10.3390/genes10050373>
- Samad, A., Zhang, J., Chen, D., Chen, X., Tucker, M., & Liang, Y. (2017). Sweet sorghum bagasse and corn stover serving as substrates for producing sophorolipids. *J Ind Microbiol Biotechnol*, 44(3), 353–362. <https://doi.org/10.1007/s10295-016-1891-y>
- Sands, R., Malcolm, S., Suttles, S., & Marshall, E. (2017). Dedicated energy crops and competition for agricultural land, U.S. In E. R. S. Agriculture (Ed.), (Economic Research Report No. 223. Economic Research Report No. 223.
- Santos, M., Rodrigues, A., & Teixeira, J. A. (2005). Production of dextran and fructose from carob pod extract and cheese whey by *Leuconostoc mesenteroides* NRRL B512 (f). *Biochemical Engineering Journal*, 25(1), 1–6. <https://doi.org/10.1016/j.bej.2005.01.022>
- Santos, L. B., Striebeck, M. V., Crespi, M. S., Capela, J. M., Ribeiro, C. A., & De Julio, M. (2016). Energy evaluation of biochar obtained from the pyrolysis of pine pellets. *Journal of Thermal Analysis and Calorimetry*, 126(3), 1879–1887. <https://doi.org/10.1007/s10973-016-5683-4>
- Saratale, G. D., & Oh, M.-K. (2015). Characterization of poly-3-hydroxybutyrate (PHB) produced from *Ralstonia eutropha* using an alkali-pretreated biomass feedstock. *International Journal of Biological Macromolecules*, 80, 627–635. <https://doi.org/10.1016/j.ijbiomac.2015.07.034>
- Saravanamurugan, S., Li, H., Riisager, A., & Pandey, A. (2020). *Biomass, biofuels, biochemicals: Recent advances in development of platform chemicals*. Elsevier.
- Sauer, M., Porro, D., Mattanovich, D., & Branduardi, P. (2008). Microbial production of organic acids: Expanding the markets. *Trends in Biotechnology*, 26(2), 100–108. <https://doi.org/10.1016/j.tibtech.2007.11.006>
- Sawisit, A., Jantama, S. S., Kanchanatawee, S., & Jantama, K. (2015). Efficient utilization of cassava pulp for succinate production by metabolically engineered *Escherichia coli* KJ122. *Bioprocess and Biosystems Engineering*, 38(1), 175–187. <https://doi.org/10.1007/s00449-014-1257-7>
- Schieber, A., Stintzing, F. C., & Carle, R. (2001). By-products of plant food processing as a source of functional compounds — recent developments. *Trends in Food Science & Technology*, 12(11), 401–413. [https://doi.org/10.1016/S0924-2244\(02\)00012-2](https://doi.org/10.1016/S0924-2244(02)00012-2)
- Schneider, D. R., Kirac, M., & Hublin, A. (2012). Cost-effectiveness of GHG emission reduction measures and energy recovery from municipal waste in Croatia. *Energy*, 48(1), 203–211. <https://doi.org/10.1016/j.energy.2012.02.008>

- Schramm, L. L., Stasiuk, E. N., & Marangoni, D. G. (2003). 2 Surfactants and their applications. *Annual Reports Section "C" (Physical Chemistry)*, 99, 3–48. <https://doi.org/10.1039/B208499F>
- Schuster, B. G., & Chinn, M. S. (2013). Consolidated bioprocessing of lignocellulosic feedstocks for ethanol fuel production. *BioEnergy Research*, 6(2), 416–435. <https://doi.org/10.1007/s12155-012-9278-z>
- Seibold, G., Auchter, M., Berens, S., Kalinowski, J., & Eikmanns, B. J. (2006). Utilization of soluble starch by a recombinant *Corynebacterium glutamicum* strain: Growth and lysine production. *Journal of Biotechnology*, 124(2), 381–391. <https://doi.org/10.1016/j.jbiotec.2005.12.027>
- Shafiee, S., & Topal, E. (2009). When will fossil fuel reserves be diminished? *Energy Policy*, 37(1), 181–189. <https://doi.org/10.1016/j.enpol.2008.08.016>
- Shahab, R. L., Luterbacher, J. S., Brethauer, S., & Studer, M. H. (2018). Consolidated bioprocessing of lignocellulosic biomass to lactic acid by a synthetic fungal-bacterial consortium. *Biotechnology and Bioengineering*, 115(5), 1207–1215. <https://doi.org/10.1002/bit.26541>
- Shapiro, R. S., Chavez, A., & Collins, J. J. (2018). CRISPR-based genomic tools for the manipulation of genetically intractable microorganisms. *Nature Reviews. Microbiology*, 16(6), 333–339. <https://doi.org/10.1038/s41579-018-0002-7>
- Sharma, R. V., Somidi, A. K. R., & Dalai, A. K. (2015). Preparation and properties evaluation of biolubricants derived from canola oil and canola biodiesel. *Journal of Agricultural and Food Chemistry*, 63(12), 3235–3242. <https://doi.org/10.1021/jf505825k>
- Shen, N., Wang, Q., Zhu, J., Qin, Y., Liao, S., Li, Y., Zhu, Q., Jin, Y., Du, L., & Huang, R. (2016). Succinic acid production from duckweed (*Landoltia punctata*) hydrolysate by batch fermentation of *Actinobacillus succinogenes* GXAS137. *Bioresource Technology*, 211, 307–312. <https://doi.org/10.1016/j.biortech.2016.03.036>
- Shi, S., Kang, L., & Lee, Y. (2015). Production of lactic acid from the mixture of softwood pre-hydrolysate and paper mill sludge by simultaneous saccharification and fermentation. *Applied Biochemistry and Biotechnology*, 175(5), 2741–2754. <https://doi.org/10.1007/s12010-014-1451-8>
- Sikarwar, V. S., Zhao, M., Clough, P., Yao, J., Zhong, X., Memon, M. Z., Shah, N., Anthony, E. J., & Fennell, P. S. (2016). An overview of advances in biomass gasification. *Energy & Environmental Science*, 9(10), 2939–2977. <https://doi.org/10.1039/C6EE00935B>
- Sikarwar, V. S., Zhao, M., Fennell, P. S., Shah, N., & Anthony, E. J. (2017). Progress in bio-fuel production from gasification. *Progress in Energy and Combustion Science*, 61, 189–248. <https://doi.org/10.1016/j.pecs.2017.04.001>
- Sirajunnisa, A. R., & Surendhiran, D. (2016). Algae—a quintessential and positive resource of bioethanol production: A comprehensive review. *Renewable and Sustainable Energy Reviews*, 66, 248–267. <https://doi.org/10.1016/j.rser.2016.07.024>
- Soccol, C. R., Faraco, V., Karp, S. G., Vandenberghe, L. P. S., Thomaz-Soccol, V., Woiciechowski, A. L., & Pandey, A. (2019). Lignocellulosic bioethanol: Current status and future perspectives. In A. Pandey. (Ed.), *Biofuels: Alternative feedstocks and conversion processes for the production of liquid and gaseous biofuels* (pp. 331–354). Academic Press.
- Song, C. W., Park, J. M., Chung, S. C., Lee, S. Y., & Song, H. (2019). Microbial production of 2,3-butanediol for industrial applications. *Journal of Industrial Microbiology & Biotechnology*, 46(11), 1583–1601. <https://doi.org/10.1007/s10295-019-02231-0>
- Sorokulova, I., Krumnow, A., Globa, L., & Vodyanoy, V. (2009). Efficient decomposition of shrimp shell waste using *Bacillus cereus* and *Exiguobacterium acetylicum*. *Journal of*

- Industrial Microbiology & Biotechnology*, 36(8), 1123–1126. <https://doi.org/10.1007/s10295-009-0587-y>
- Sosulski, K., Wang, S. M., Ingledew, W. M., Sosulski, F. W., & Tang, J. (1997). Preprocessed barley, rye, and triticale as a feedstock for an integrated fuel ethanol feedlot plant. *Applied Microbiology and Biotechnology*, 63(5), 59–70.
- Speirs, J., McGlade, C., & Slade, R. (2015). Uncertainty in the availability of natural resources: Fossil fuels, critical metals and biomass. *Energy Policy*, 87, 654–664. <https://doi.org/10.1016/j.enpol.2015.02.031>
- Stelte, W. (2012). Guideline: Storage and handling of wood pellets. Danish Technological Institute.
- Su, B., Wu, M., Zhang, Z., Lin, J., & Yang, L. (2015). Efficient production of xylitol from hemicellulosic hydrolysate using engineered *Escherichia coli*. *Metabolic Engineering*, 31, 112–122. <https://doi.org/10.1016/j.ymben.2015.07.003>
- Takkellapati, S., Li, T., & Gonzalez, M. A. (2018). An overview of biorefinery derived platform chemicals from a cellulose and hemicellulose biorefinery. *Clean Technologies and Environmental Policy*, 20(7), 1615–1630. <https://doi.org/10.1007/s10098-018-1568-5>
- Takumi, K., Ziyatdinov, M. K., Samsonov, V., & Nonaka, G. (2017). Fermentative production of cysteine by *Pantoea ananatis*. *Applied and Environmental Microbiology*, 83(5), e02502–02516. <https://doi.org/10.1128/AEM.02502-16>
- Talebnaia, F., Karakashev, D., & Angelidaki, I. (2010). Production of bioethanol from wheat straw: An overview on pretreatment, hydrolysis and fermentation. *Bioresource Technology*, 101(13), 4744–4753. <https://doi.org/10.1016/j.biortech.2009.11.080>
- Tanger, P., Field, J. L., Jahn, C. E., Defoort, M. W., & Leach, J. E. (2013). Biomass for thermochemical conversion: Targets and challenges. *Frontiers in Plant Science*, 4, 218–218. <https://doi.org/10.3389/fpls.2013.00218>
- Taskin, M., & Erdal, S. (2011). Production of carotenoids by *Rhodotorula glutinis* MT-5 in submerged fermentation using the extract from waste loquat kernels as substrate. *Journal of the Science of Food and Agriculture*, 91(8), 1440–1445. <https://doi.org/10.1002/jsfa.4329>
- Tateno, T., Okada, Y., Tsuchidate, T., Tanaka, T., Fukuda, H., & Kondo, A. (2009). Direct production of cadaverine from soluble starch using *Corynebacterium glutamicum* coexpressing alpha-amylase and lysine decarboxylase. *Applied Microbiology and Biotechnology*, 82(1), 115–121. <https://doi.org/10.1007/s00253-008-1751-4>
- Taylor, R., Natrass, L., Alberts, G., Robson, P., Chudziak, C., Bauen, A., Libelli, I., Lotti, G., Prussi, M., & Nistri, R. (2015). From the sugar platform to biofuels and biochemicals: Final report for the European commission directorate-general energy.
- Teekens, A. M., Bruins, M. E., van Kasteren, J. M., Hendriks, W. H., & Sanders, J. P. (2016). Synergy between bio-based industry and the feed industry through biorefinery. *Journal of the Science of Food and Agriculture*, 96(8), 2603–2612. <https://doi.org/10.1002/jsfa.7596>
- Teong, S. P., Yi, G., & Zhang, Y. (2014). Hydroxymethylfurfural production from biore-sources: Past, present and future. *Green Chemistry*, 16(4), 2015–2026. <https://doi.org/10.1039/c3gc42018c>
- Thavasi, R., Jayalakshmi, S., & Banat, I. M. (2011). Application of biosurfactant produced from peanut oil cake by *Lactobacillus delbrueckii* in biodegradation of crude oil. *Bioresource Technology*, 102(3), 3366–3372. <https://doi.org/10.1016/j.biortech.2010.11.071>
- Thavasi, R., Nambaru, V. S., Jayalakshmi, S., Balasubramanian, T., & Banat, I. M. (2011). Biosurfactant production by *Pseudomonas aeruginosa* from renewable resources. *Indian Journal of Microbiology*, 51(1), 30–36. <https://doi.org/10.1007/s12088-011-0076-7>

- Thiffault, E., Barrette, J., Blanchet, P., Nguyen, Q. N., & Adjalle, K. (2019). Optimizing quality of wood pellets made of hardwood processing residues. *Forests*, 10, 607. <https://doi.org/10.3390/f10070607>
- Thomas, A. O., Leahy, M. C., Smith, J. W. N., & Spence, M. J. (2017). Natural attenuation of fatty acid methyl esters (FAME) in soil and groundwater. *Quarterly Journal of Engineering Geology and Hydrogeology*, 50(3), 301–317. <https://doi.org/10.1144/qjegh2016-130>
- Thuy, N. T. H., Kongkaew, A., Flood, A., & Boontawan, A. (2017). Fermentation and crystallization of succinic acid from *Actinobacillus succinogenes* ATCC55618 using fresh cassava root as the main substrate. *Bioresource Technology*, 233, 342–352. <https://doi.org/10.1016/j.biortech.2017.02.114>
- Tian, L., Conway, P. M., Cervenka, N. D., Cui, J., Maloney, M., Olson, D. G., & Lynd, L. R. (2019). Metabolic engineering of *Clostridium thermocellum* for n-butanol production from cellulose. *Biotechnology for Biofuels*, 12, 186 <https://doi.org/10.1186/s13068-019-1524-6>
- Tinoi, J., Rakariyatham, N., & Deming, R. (2006). Utilization of mustard waste isolates for improved production of astaxanthin by *Xanthophyllomyces dendrorhous*. *J Ind Microbiol Biotechnol*, 33(4), 309–314. <https://doi.org/10.1007/s10295-005-0054-3>
- Tripathi, N., Hills, C. D., Singh, R. S., & Atkinson, C. J. (2019). Biomass waste utilisation in low-carbon products: Harnessing a major potential resource. *Npj Climate and Atmospheric Science*, 2(1), 35. <https://doi.org/10.1038/s41612-019-0093-5>
- Tsang, Y. F., Kumar, V., Samadar, P., Yang, Y., Lee, J., Ok, Y. S., Song, H., Kim, K.-H., Kwon, E. E., & Jeon, Y. J. (2019). Production of bioplastic through food waste valorization. *Environment International*, 127, 625–644. <https://doi.org/10.1016/j.envint.2019.03.076>
- Tumuluru, J. S., Sokhansanj, S., Hess, J. R., Wright, C. T., & Boardman, R. D. (2011). A review on biomass torrefaction process and product properties for energy applications. *Industrial Biotechnology*, 7(5), 384–401. <https://doi.org/10.1089/ind.2011.7.384>
- Unrean, P., & Ketsub, N. (2018). Integrated lignocellulosic bioprocess for co-production of ethanol and xylitol from sugarcane bagasse. *Industrial Crops and Products*, 123, 238–246. <https://doi.org/10.1016/j.indcrop.2018.06.071>
- Uslu, A., Faaij, A. P., & Bergman, P. C. (2008). Pre-treatment technologies, and their effect on international bioenergy supply chain logistics. Techno-economic evaluation of torrefaction, fast pyrolysis and pelletisation. *Energy*, 33(8), 1206–1223. <https://doi.org/10.1016/j.energy.2008.03.007>
- Varzakas, T., Kandyli, P., Dimitrellou, D., Salamoura, C., Zakynthinos, G., & Proestos, C. (2018). 6 - Innovative and fortified food: Probiotics, prebiotics, GMOs, and superfood. In M. E. Ali and N. N. A. Nizar (Eds.), *Preparation and processing of religious and cultural foods* (pp. 67–129). Woodhead Publishing.
- Vedaraman, N., & Venkatesh, N. (2011). Production of surfactin by *Bacillus subtilis* MTCC 2423 from waste frying oils. *Brazilian Journal of Chemical Engineering*, 28(2), 175–180. <https://doi.org/10.1590/S0104-66322011000200001>
- Vedyashkina, T., Revin, V., & Gogotov, I. (2005). Optimizing the conditions of dextran synthesis by the bacterium *Leuconostoc mesenteroides* grown in a molasses-containing medium. *Applied Biochemistry and Microbiology*, 41(4), 361–364. <https://doi.org/10.1007/s10438-005-0061-1>
- Verduyck, J., & De Vos, D. E. (2017). Controlled defunctionalisation of biobased organic acids. *Chemical Communications (Cambridge, England)*, 53(42), 5682–5693. <https://doi.org/10.1039/c7cc01380a>

- Viell, J., Harwardt, A., Seiler, J., & Marquardt, W. (2013). Is biomass fractionation by Organosolv-like processes economically viable? A conceptual design study. *Bioresource Technology*, 150, 89–97. <https://doi.org/10.1016/j.biortech.2013.09.078>
- Vom Stein, T., Grande, P. M., Kayser, H., Sibilla, F., Leitner, W., & de María, P. D. (2011). From biomass to feedstock: One-step fractionation of lignocellulose components by the selective organic acid-catalyzed depolymerization of hemicellulose in a biphasic system. *Green Chemistry*, 13(7), 1772–1777. <https://doi.org/10.1039/c1gc00002k>
- Walker, S., & Rothman, R. (2020). Life cycle assessment of bio-based and fossil-based plastic: A review. *Journal of Cleaner Production*, 261, 121158. <https://doi.org/10.1016/j.jclepro.2020.121158>
- Wang, M., Dewil, R., Maniatis, K., Wheeldon, J., Tan, T., Baeyens, J., & Fang, Y. (2019). Biomass-derived aviation fuels: Challenges and perspective. *Progress in Energy and Combustion Science*, 74, 31–49. <https://doi.org/10.1016/j.pecs.2019.04.004>
- Wang, Z., Li, H., Feng, J., Zhang, A., Ying, H., He, X., Jiang, M., Chen, K., & Ouyang, P. (2018). Enhanced succinic acid production from polyacrylamide-pretreated cane molasses in microbial electrolysis cells. *Journal of Chemical Technology & Biotechnology*, 93(3), 855–860. <https://doi.org/10.1002/jctb.5440>
- Wang, H., Li, L., Zhang, L., An, J., Cheng, H., & Deng, Z. (2016). Xylitol production from waste xylose mother liquor containing miscellaneous sugars and inhibitors: one-pot biotransformation by *Candida tropicalis* and recombinant *Bacillus subtilis*. *Microb Cell Fact*, 15, 82 <https://doi.org/10.1186/s12934-016-0480-0>
- Wang, Y., Qiu, L., Zhu, M., Sun, G., Zhang, T., & Kang, K. (2019). Comparative evaluation of hydrothermal carbonization and low temperature pyrolysis of *Eucommia ulmoides* olive for the production of solid biofuel. *Scientific Reports*, 9(1), 5535 <https://doi.org/10.1038/s41598-019-38849-4>
- Wang, A., Xu, Y., Ma, C., Gao, C., Li, L., Wang, Y., Tao, F., & Xu, P. (2012). Efficient 2,3-butanediol production from cassava powder by a crop-biomass-utilizer, *Enterobacter cloacae* subsp. *dissolvens* SDM. *Plos One*, 7(7), e40442 <https://doi.org/10.1371/journal.pone.0040442>
- Wang, P., Yu, H., Zhan, S., & Wang, S. (2011). Catalytic hydrolysis of lignocellulosic biomass into 5-hydroxymethylfurfural in ionic liquid. *Bioresource Technology*, 102(5), 4179–4183. <https://doi.org/10.1016/j.biortech.2010.12.073>
- Wargacki, A. J., Leonard, E., Win, M. N., Regitsky, D. D., Santos, C. N. S., Kim, P. B., Cooper, S. R., Raisner, R. M., Herman, A., Sivitz, A. B., Lakshmanaswamy, A., Kashiyama, Y., Baker, D., & Yoshikuni, Y. (2012). An engineered microbial platform for direct biofuel production from brown macroalgae. *Science (New York, N.Y.)*, 335(6066), 308–313. <https://doi.org/10.1126/science.1214547>
- Wärtsilä (2014). *Biogas liquefaction plant supplied by Wärtsilä to produce biofuel for buses in Norway*. <https://www.wartsila.com/media/news/12-02-2014-biogas-liquefaction-plant-supplied-by-wartsila-to-produce-biofuel-for-buses-in-norway>
- WBA (2018). *WBA global bionergy statistics 2018*. https://worldbioenergy.org/uploads/181203%20WBA%20GBS%202018_hq.pdf
- Welker, C. M., Balasubramanian, V. K., Petti, C., Rai, K. M., DeBolt, S., & Mendu, V. (2015). Engineering plant biomass lignin content and composition for biofuels and bioproducts. *Energies*, 8(8), 7654–7676. <https://doi.org/10.3390/en8087654>
- Werpy, T., & Petersen, G. (2004). Top value added chemicals from biomass: Volume I—results of screening for potential candidates from sugars and synthesis gas.

- Wilkens, E., Ringel, A. K., Hortig, D., Willke, T., & Vorlop, K.-D. (2012). High-level production of 1,3-propanediol from crude glycerol by *Clostridium butyricum* AKR102a. *Appl Microbiol Biotechnol*, 93(3), 1057–1063. <https://doi.org/10.1007/s00253-011-3595-6>
- Wisjnuprpto, N. A., Helmy, Q., Kardena, E., & Funamizu, N. (2011). Strategies toward commercial scale of biosurfactant production as potential substitute for it's chemically counterparts. *International Journal of Biotechnology*, 12(1/2), 66–86. <https://doi.org/10.1504/IJBT.2011.042682>
- Xiao, Z., Yuan, X., Jiang, L., Chen, X., Li, H., Zeng, G., Leng, L., Wang, H., & Huang, H. (2015). Energy recovery and secondary pollutant emission from the combustion of co-pelletized fuel from municipal sewage sludge and wood sawdust. *Energy*, 91, 441–450. <https://doi.org/10.1016/j.energy.2015.08.077>
- Xing, X., Fan, F., & Jiang, W. (2018). Characteristics of biochar pellets from corn straw under different pyrolysis temperatures. *Royal Society Open Science*, 5(8), 172346 <https://doi.org/10.1098/rsos.172346>
- Xu, F., Li, Y., Ge, X., Yang, L., & Li, Y. (2018). Anaerobic digestion of food waste - Challenges and opportunities. *Bioresource Technology*, 247, 1047–1058. <https://doi.org/10.1016/j.biortech.2017.09.020>
- Xu, Q., Singh, A., & Himmel, M. E. (2009). Perspectives and new directions for the production of bioethanol using consolidated bioprocessing of lignocellulose. *Current Opinion in Biotechnology*, 20(3), 364–371. <https://doi.org/10.1016/j.copbio.2009.05.006>
- Yamane, T., & Tanaka, R. (2013). Highly accumulative production of l(+)-lactate from glucose by crystallization fermentation with immobilized *Rhizopus oryzae*. *Journal of Bioscience and Bioengineering*, 115(1), 90–95. <https://doi.org/10.1016/j.jbiosc.2012.08.005>
- Yang, S., Mohagheghi, A., Franden, M. A., Chou, Y.-C., Chen, X., Dowe, N., Himmel, M. E., & Zhang, M. (2016). Metabolic engineering of *Zymomonas mobilis* for 2,3-butanediol production from lignocellulosic biomass sugars. *Biotechnology for Biofuels*, 9(1), 189 <https://doi.org/10.1186/s13068-016-0606-y>
- Yang, W., Shimanouchi, T., Iwamura, M., Takahashi, Y., Mano, R., Takashima, K., Tanifuji, T., & Kimura, Y. (2015). Elevating the fuel properties of *Humulus lupulus*, *Plumeria alba* and *Calophyllum inophyllum* L. through wet torrefaction. *Fuel*, 146, 88–94. <https://doi.org/10.1016/j.fuel.2015.01.005>
- Yang, X., Xu, M., & Yang, S.-T. (2015). Metabolic and process engineering of *Clostridium cellulovorans* for biofuel production from cellulose. *Metabolic Engineering*, 32, 39–48. <https://doi.org/10.1016/j.ymben.2015.09.001>
- Yeo, I., Lee, Y. J., Song, K., Jin, H. S., Lee, J. E., Kim, D., Lee, D. W., & Kang, N. J. (2018). Low-molecular weight keratins with anti-skin aging activity produced by anaerobic digestion of poultry feathers with *Fervidobacterium islandicum* AW-1. *Journal of Biotechnology*, 271, 17–25. <https://doi.org/10.1016/j.jbiotec.2018.02.003>
- Younes, I., & Rinaudo, M. (2015). Chitin and chitosan preparation from marine sources. Structure, properties and applications. *Marine Drugs*, 13(3), 1133–1174. <https://doi.org/10.3390/md13031133>
- Yu, K. O., Jung, J., Kim, S. W., Park, C. H., & Han, S. O. (2012). Synthesis of FAEEs from glycerol in engineered *Saccharomyces cerevisiae* using endogenously produced ethanol by heterologous expression of an unspecific bacterial acyltransferase. *Biotechnology and Bioengineering*, 109(1), 110–115. <https://doi.org/10.1002/bit.23311>
- Yu, J., & Si, Y. (2004). Metabolic carbon fluxes and biosynthesis of polyhydroxyalkanoates in *Ralstonia eutropha* on short chain fatty acids. *Biotechnology Progress*, 20(4), 1015–1024. <https://doi.org/10.1021/bp034380e>

- Zahan, K. A., & Kano, M. (2018). Biodiesel production from palm oil, its by-products, and mill effluent: A review. *Energies*, 11(8), 2132. <https://doi.org/10.3390/en11082132>
- Zahed, O., Jouzani, G. S., Abbasalizadeh, S., Khodaiyan, F., & Tabatabaei, M. (2016). Continuous co-production of ethanol and xylitol from rice straw hydrolysate in a membrane bioreactor. *Folia Microbiologica*, 61(3), 179–189. <https://doi.org/10.1007/s12223-015-0420-0>
- Zang, G., Shah, A., & Wan, C. (2020). Techno-economic analysis of an integrated biorefinery strategy based on one-pot biomass fractionation and furfural production. *Journal of Cleaner Production*, 260, 120837. <https://doi.org/10.1016/j.jclepro.2020.120837>
- Zhang, J., Geng, A., Yao, C., Lu, Y., & Li, Q. (2012). Xylitol production from D-xylose and horticultural waste hemicellulosic hydrolysate by a new isolate of *Candida athensensis* SB18. *Bioresource Technology*, 105, 134–141. <https://doi.org/10.1016/j.biortech.2011.11.119>
- Zhang, J., Loh, K. C., Lee, J., Wang, C. H., Dai, Y., & Wah Tong, Y. (2017). Three-stage anaerobic co-digestion of food waste and horse manure. *Scientific Reports*, 7(1), 1269 <https://doi.org/10.1038/s41598-017-01408-w>
- Zhang, Z.-b., Lu, Q., Ye, X.-n., Wang, T.-p., Wang, X.-h., & Dong, C.-q. (2015). Selective production of levoglucosenone from catalytic fast pyrolysis of biomass mechanically mixed with solid phosphoric acid catalysts. *BioEnergy Research*, 8(3), 1263–1274. <https://doi.org/10.1007/s12155-015-9581-6>
- Zhang, F., Rodriguez, S., & Keasling, J. D. (2011). Metabolic engineering of microbial pathways for advanced biofuels production. *Curr Opin Biotechnol*, 22(6), 775–783. <https://doi.org/10.1016/j.copbio.2011.04.024>
- Zhang, X., Xiao, N., Chen, M., Wei, Y., & Liu, C. (2020). Functional packaging films originating from hemicelluloses laurate by direct transesterification in ionic liquid. *Carbohydrate Polymers*, 229, 115336 <https://doi.org/10.1016/j.carbpol.2019.115336>
- Zhang, L., Zhao, R., Jia, D., Jiang, W., & Gu, Y. (2020). Engineering *Clostridium ljungdahlii* as the gas-fermenting cell factory for the production of biofuels and biochemicals. *Curr Opin Chem Biol*, 59, 54–61. <https://doi.org/10.1016/j.cbpa.2020.04.010>
- Zhou, N., Zhang, Y., Wu, X., Gong, X., & Wang, Q. (2011). Hydrolysis of Chlorella biomass for fermentable sugars in the presence of HCl and MgCl₂. *Bioresource Technology*, 102(21), 10158–10161. <https://doi.org/10.1016/j.biortech.2011.08.051>